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A Machine Learning-Based Forecasting Approaches and Correlation Analysis to Assess Future Extreme Heat Scenarios in Bangladesh and Its Impact on Public Health

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ABSTRACT

Bangladesh's geographical location as well as dense population makes it highly vulnerable to rising maximum temperatures and the associated public health impacts driven by climate change. Although machine learning-based forecasting and correlation analysis are widely applied, most existing models lack regional calibration and consequently fail to capture Bangladesh's distinct climatic variability. In addition, the accuracy of forecasting of maximum temperature cannot be determined since the results are inconsistent. This study addresses these challenges by proposing and evaluating four forecasting models: the Autoregressive Integrated Moving Average (ARIMA) model, the Exponential Smoothing State-Space (ETS) model, the Holt (Double Exponential Smoothing) model, and the Prophet model. The models forecast maximum temperature and examine its correlation with public health issues—particularly diarrhea cases in Bangladesh—over the period from 2017 to 2040. The study compares the forecasting performance of these state-of-the-art models, showing that maximum temperatures are projected to range between 31.39°C and 34.3°C by 2040. ETS and Holt predicted 3786 and 2282 diarrhea cases respectively at 33.71°C, while Prophet estimated 3841 cases at 34.3°C. ARIMA forecasted 4045 diarrhea cases with a predicted maximum temperature of 32.15°C, achieving a moderate error margin. Among the models, ARIMA demonstrated the most balanced and reliable performance, confirming its effectiveness for climate and health forecasting in the Bangladeshi context. Overall, this work outperforms existing approaches by delivering consistent and accurate temperature forecasts. Additionally, it bridges a significant gap in the literature by establishing a clear correlation between maximum temperature and diarrhea cases in Bangladesh. The findings offer actionable insights for policymakers and public health officials, supporting more effective strategies for public health management and climate adaptation planning.

1 | Introduction

In the present context, the increase in air temperatures due to climate change is causing an uptick in more frequent heat waves across the globe that are affecting people's health, ecosystems, and economies all across the world [1]. The expectation is that global warming will increase the occurrence of heat waves frequently in the future; recent progress in using data driven forecasting methods, such as deep learning models, has notably improved the precision of predicting long term heatwaves [2].

The data gathered by the Bangladesh Meteorological Department (BMD) underscores the significance of forecasting relative humidity levels due to its role in enhancing the precision of weather predictions and supporting sectors such as agriculture and public health while also improving daily comfort levels and preparedness for severe weather occurrences, such as storms or heat waves [3]. Heat waves do represent a substantial threat to sustainable development and human safety, significantly increasing risks such as health effects on the public in general [4]. In 2023, Bangladesh faced some of the hottest-ever summer temperatures in South Asia. During the April to July hot season, average temperatures hover around 28°C, with typical highs climbing between ranges of about 30°C and 36°C. By 2023, temperatures were noticeably higher, consistently above 35°C and sometimes edging beyond 40°C [4]. The field of climate change science is manifold of AI and ML techniques, which are widely utilized to disseminate the literature and enhance knowledge about the impact of climate change on health for handling Big Data [5].

Concerning to the climate change, the necessity to forecast the further weather conditions particularly temperatures to prevent from climate change conditions, as well as to provide advices and directions how to address emergent climate change conditions is crucial [6]. The establishment of effective heatwave action plans requires monitoring, tracking, and prediction of heat waves [7]. Particularly in Bangladesh during the pre-monsoon season, heat waves—which are characterized by extreme temperatures and high humidity—pose significant risks to health and result in illness [8]. Additionally, the higher land surface temperatures in urban areas decrease human thermal comfort worsen living conditions and increase hazards to ecosystems [9]. Therefore, studying and solving global challenges by predicting their future changes is difficult, and time series are essential tools for climate data analysis, especially temperature, which has been considered as the principal climatic parameter [10]. On the other hand, this distribution of heat waves and associated extreme high temperatures are responsible for their being disastrous across the country from a health, economic, agricultural as well as an ecological perspective in Bangladesh [11]. In addition, the present global rate of carbon emissions due to the combustion energy consumption with fossil fuels and anthropogenic activities are more rapid than at any time in human history, resulting in an imbalance CO_2 budget, which is one of the contributing factors for causing planet warming since air and ocean temperatures began rapidly rising hundred years ago [12]. Artificial Intelligence (AI) and Machine Learning (ML) techniques illustrate revolutionizing climate change research by enhancing understanding of its impact on human health and expanding the scope of literature. The problem with many models is inadequate data, a lack of consistency, and uneven coverage of the world, which demonstrates that the technologies still require further development before all the benefits can be fully accrued.

There is a pressing need for research that not only builds on existing methods but also innovates new approaches that consider the unique climatic and socio-economic characteristics of regions in Bangladesh. Therefore, the temperature has a huge impact on climate variability and that is why it remains one of the most pressing challenges to predict with any reasonable degree of accuracy. So, this time series analysis and forecasting model is a time series significant tool to analyze variability, trends, and patterns that can be done through many models. Therefore, the paper introduced four forecasting time series models for predicting maximum temperature and diarrhea cases up to 2040, including the ETS model, Holt model, ARIMA model, and the Prophet model to utilize statistical tests such as Augmented Dickey-Fuller Test (ADF Test), Autocorrelation Function (ACF), and Partial Autocorrelation Function (PACF), confidence intervals, and evaluation error functions such as Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) precisely. Such approaches do not only ameliorate better accuracy in future forecasts of temperatures and diseases by taking trends, seasonality, and long-term trends into account, they also provide early-warning and mitigation plans by detecting any emergent risks between climate and health before they escalate into considerable threats. This work incorporates actionable insights for policymakers, public health officials, and this research contributes to better public health management and more effective climate adaptation strategies, thereby mitigating the adverse effects of extreme heat events on public health.

The rest of the article is organized as: Section 2 provides a review of the literature on work-related, followed by methodology in Section 3. The results and discussion are presented in Section 4 and followed by a conclusion in Section 5.

2 | Literature Review

This section describes the recent research papers in time series analysis and forecasting models of climate prediction concentrating its applicability to temperature variability in Bangladesh. In recent years, Bangladesh, being a tropical country with high population density and a scenario of economic vulnerability, has been one of the world's most heat wave-prone countries experiencing deadly health outcomes from extreme temperature and heat wave events influenced by climate change. Maria et al. [4], the research suggested using the vulnerability index and the most heat vulnerable districts in Bangladesh. This study shows that Kurigram is very heat sensitive; this is why it is more agriculturally involved, most of the residents are illiterate, and many people are impoverished. Sherpur is the least adaptive district because there are quite a few health workers and not many green spaces. Furthermore, Sherpur and Khagrachhari were identified as the most vulnerable districts to heatwaves. In Reference [8], the study mentions the pre-monsoon season of 2021, where heat waves were witnessed in Bangladesh. The analysis exhibits the conditions such as a thermal low over the western side, the anticyclonic circulation over the Bay of Bengal, high humidity at lower tropospheric levels, and a negative lifted index that are well known for the formation, intensification, and persistence of the heat wave in Bangladesh.

A study by Imran et al. [9] proposed that the study reveals that Dhaka city's built-up area increased by 67% from 1993 to 2020, replacing lowland with vegetation, bare soil, and water bodies. Land Surface Temperature increased by 0.24°C per year, causing human thermal discomfort from moderate to strong heat stress. In addition, the study found that vegetation and water index. This paper concluded that Dhaka city was substantially altered due to rapid urbanization and socio-economic development. Farhana Akter Bina [10] proposed the Sliced Functional

Time Series (SFTS) model for forecasting monthly average temperature from 2020 to 2039 in Bangladesh. Apart from this, this paper exhibits that the ARIMA model forecasts a monthly average temperature of 28.81°C (June), while the ETS model predicts a monthly average temperature of 28.15°C (June) in 2022. The SFTS model, yet, demonstrates a slightly higher average temperature of 29.26°C (June) in 2022. Nevertheless, the research does not correctly split the dataset into training and testing sets.

In Reference [11], the study also found upward trends of heat wave conditions in the south-west Bangladesh because of specific tropospheric conditions. The study indicates that the monthly, seasonal, and annual frequency of Maximum Temperature ($\geq$ 36°C) in Bangladesh is increasing trends, with the highest rates observed in the southwestern part of the country. The paper demonstrates the impacts of climate change on livestock health in Bangladesh, specifically increased heat stress and disease, providing suggestions to better management practices and adaptation strategies; however, there remains a gap between the systematic studies conducted with respect to the particular context of Bangladesh in [12]. Sneha et al. [13], analyze Bangladesh's temperature data over the past 100 years, revealing an average increase of 1°C. Linear regression, polynomial regression, and support vector regression (SVR) models are evolved to predict the average temperature, identifying polynomial regression as the best-performing model in Bangladesh. However, in this study, data separation for training and testing is unclear, relies on stable data, fails to consider appropriate trends, and failed to mention ADF, ACF/PACF, and confidence interval analysis.

In Sarkar et al. [14], this article also delineated the specific future groundwater potential zones for 2025, 2030, 2035, and 2040 through a machine learning technique and future RCP. The ANN model performed better in accuracy with 23.10% of the land classed under very high groundwater potential and 33.50% in extremely high potential area. In Reference [15], the researcher applies Landsat data and machine algorithms to evaluate the LULC effect on LST in Kabul City, Afghanistan. The current analysis reveals that built-up areas have expanded while vegetation and bare soil significantly decreased, and seasonal analysis exhibits higher LST during the summer. The projections indicate that built-up areas will rise to 18% and 20% in 2030 and 2040, respectively. In Reference [16], the study displays that more than 80% total area has very high Urban Heat Island (UHI) values, which can be useful in designing strategies to reduce UHI using machine learning. This article also forecasted in Reference [17], the urban green cover was projected to reduce by 13% while the built-up area was projected to rise by 21% in 2030 with the maximum increase in LST of 9.29°C. The maximum temperature is expected in the constructed and land-use category, and minimum in the vegetation cover and water body category. In Al Kafy et al. [18], the study also estimates a significant area of LULCs 25 years before the current study period again through LULCs replacement with the built-up area. The population density of the built-up areas rose from 1995 to 2015 by 16.23%, and that of agricultural areas declined by 38.88%. The maximum temperature rose from 37.22°C in 1995 to 42.7°C in summer and 22.18°C to 28.94°C in winter in 2020. This paper posits that the high variability of LULC is primarily responsible for the rising LSTs. In Reference [19], another study on heat waves in Bangladesh from 1981 to 2016 presents substantial warming trends with more frequent high-temperature days. Moreover, in 2014, the maximum temperature exceeded 36°C, primarily in April and May, for many days. In 2018, it exceeded 36°C occasionally, except in April. This research illustrates that heat waves enter Bangladesh from the west due to the advection of higher maximum temperature from the west.

However, existing knowledge is limited in scope, as no large-scale study has been undertaken to systematically explore the challenges introduced by Bangladesh's tropical environment and its dense population with wider economic vulnerabilities. In existing research paper gap lies those recent studies do not mention clearly regarding splitting dataset into training and testing, not rely on stable data, fail to consider appropriate trends, do not include significant statistical tests such as the ADF test, ACF/PACF analysis or confidence intervals techniques. Earlier paper focused on the monthly average temperature, does not focus the maximum temperature, and fail to capture the appropriate trajectory over the time period. Moreover, there is no perfect relevant research for health diseases such as diarrhea cases that addresses the specific challenges posed by Bangladesh's tropical climate. Thus, more research needs to be done in assessing the generalizability and reliability of these models. Therefore, the proposed outcomes and findings together exhibit improvement concerning state-of-the-art solutions. Therefore, the proposed forecast provides policymakers and public health officials with real-time strategies to ameliorate heat-health planning in susceptible areas, which can reduce the adverse effects of hotter events on human beings.

The assessed study exhibits major gaps in the existing research and raises crucial questions which are stated below:

2.1 | Research Questions

I. What is the impact of modeling precise trends? How do advanced time series models (e.g., Prophet) perform compared to traditional approaches (e.g., ARIMA, polynomial regression) in predicting maximum temperature?

II. How does incorporating statistical techniques (e.g., ADF test, ACF/PACF, and confidence intervals) ameliorate the performance and interpretability of temperature forecasting models in Bangladesh?

III. How poses a health risk to the population, especially in terms of heat-related diseases?

The objective of the work provides the use of improved statistical analysis and a new approach in pursuit of more accurate and reliable results in the context of the research question of the existing paper.

2.2 | Research Objectives

i. This article proposes viable forecasting approaches for extreme heat in Bangladesh using machine learning based on the ETS, ARIMA, Holt, and Prophet Models, with a research investigation of the potential health implications.

ii. The novelty of the proposed scheme is that the study exhibits appropriateness for forecasting maximum temperatures up to 2040 that correlates extreme heat events with heat-related diseases such as diarrhea cases to predict future public health scenarios up to 2040.

iii. To ameliorate the reliability of forecasting temperatures and diarrhea cases in Bangladesh, utilize statistical tests (ADF, ACF/PACF, confidence intervals) and evaluation error function (RMSE, MAE) to handle models.

The key contributions of this work are evolving a climate-health forecasting framework, which combines both statistical validation and time-series models to generate more precise and actionable information about the correlation between extreme heat and diarrhea cases in Bangladesh.

2.3 | Research Contributions

i. This paper demonstrates that maximum temperature is statistically significantly associated with the number of cases of diarrhea in Bangladesh, and machine learning models (ETS, ARIMA, Holt, and Prophet) can successfully predict climate-health interactions until 2040, with ARIMA being most effective.

ii. The research presents a new forecasting model to appropriate the superior accuracy, reliability, and pragmatic implications to future climate health planning by utilizing better statistical analysis (ADF, ACF/PACF, Pearson correlation, and regression) and metrics of evaluation (RMSE, MAE).

3 | Methodology

This section covers the following: ETS Model, Holt Model, ARIMA Model, Prophet Model, Monte Carlo with confidence intervals, regression analysis, correlation analysis, and methodology for forecast analysis of maximum temperature and Diarrhea cases.

3.1 | Forecasting Approach

This research work is based on making forecasts of maximum temperatures by using different time series analysis techniques such as the ETS Model, Holt Model, ARIMA Model and Prophet Model. These processes are used to examine past data and predict what future changes in temperature may be able to reveal regarding climate characteristics.

3.1.1 | Exponential Smoothing State-Space (ETS) Model

ETS model considers the errors, trend, and seasonal elements of a particular time series and tests the potential alternative models; then the best performing model is picked to simulate the data [20]. The primary focus area of the ETS model is for univariate time series forecasting, whose components are primarily trend and season. This model is non-stationary and can provide seasonal components that have various attributes and trends. Three parameters make up this model: error, trend, and seasonal. Each parameter has four possible values: A stands for additive; is a trend and Ad for additive damped is a subtype of trend, M for multiplicative, N for none, and Z for automatic selection in specific software implementations. Consequently, ETS (A, M, N) stands—Additive for Error, Multiplicative for Trend, and none for seasonal component [21]. This systematic procedure enables the ETS model to flexibly accommodate various trends in the data to enhance the accuracy and understandability of predictions of univariate time series.

3.1.2 | Holt (Double Exponential) Model

One popular method for forecasting time-series data is the Holt method, sometimes known as the Holt's method [22]. In the present study, this approach was utilized to anticipate the maximum temperature and demonstrate certain trends. The following can be used to determine the expressed value of t periods:

$$F_{n+t} = C_n + tM_n \tag{1}$$

where C_n represents the forecast intercept; M_n represents the forecast slope; F_{n+t} represents the forecast value in period t.

3.1.3 | Autoregressive Integrated Moving Average (ARIMA) Model

One of the statistical models for forecasting and analyzing time series data is this one. If $\{e_t\}$ represents white noise with zero mean and variance, then a process of order q is referred to be a moving average (MA) process [21]. The following equation should be represented as:

$$Y_t = e_t + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \dots + \beta_q e_{t-q} \tag{2}$$

When the process is described as an auto-regressive (AR) process of order, it is referred to as follows:

$$Y_t = \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \dots + \alpha_p Y_{t-p} + e_t \tag{3}$$

The combined model and models are known as ARMA models. This model is defined as.

$$Y_t = \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \dots + \alpha_p Y_{t-p} + e_t + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \dots + \beta_q e_{t-q} \tag{4}$$

where for every, it is considered that e_t is independent of. If the difference of a time series is stationary then it is said to be an ARIMA model. A process is defined as $\{Z_t\}$ following a model. An example of a model is,

$$Z_t = \alpha_1 Z_{t-1} + \alpha_2 Z_{t-2} + \dots + \alpha_p Z_{t-p} + e_t + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \dots + \beta_q e_{t-q} \tag{5}$$

3.1.4 | Prophet Model

Prophet model is developed by Facebook as an open-source tool to predict patterns in data showing seasonality s(t), holidays h(t), or changing trends g(t). It consists as [23]:

$$y(t) = g(t) + \mathrm{s}(t) + h(t) + \varepsilon(t) \tag{6}$$

This approach offers a wide range of useful amenities. Since this sector follows a pattern with both week-to-week and year-to-year events, seasonal component $s(t)$ is a flexible model to use for periodic changes. The $h(t)$ in the function captures the regular changes. Those included in annual abnormal days are holidays, less normal days, and days listed as irregular schedules. The error term, $\varepsilon(t)$ reflects information not expressed in the model.

3.2 | Monte Carlo With Confidence Intervals (CIs)

The Monte Carlo (MC) approach provides sampling distributions and confidence intervals (CIs) of statistics that are a function of other statistics. It accomplishes this based on point estimates of the component statistics, the sampling variance–covariance matrix of the component statistics, and a specified parametric distribution of those component statistics. This technique may be outlined as a procedure consisting of two steps. The initial step is called the estimation step where the parameters and their sampling variance–covariance matrix are estimated. The second step is known as the simulation step, whereby these estimates are used to draw simulated values using the specified distribution [24]. The Monte Carlo simulation step is associated as.

$$\hat{\theta}^{*} \sim \mathcal{N}\left(\mu = \hat{\theta}, \sum = \widehat{V}(\hat{\theta})\right) \tag{7}$$

Each symbol represents:

- $\hat{\theta}^{*}$: A simulated draw of the distribution of parameter vector.
- $\mathcal{N}\left(\mu, \sum\right)$: A multivariate normal distribution with:
  - Mean vector, $\mu = \hat{\theta}$: The estimated parameters from the estimation step.
  - $\widehat{V}\ (\hat{\theta})$: The estimated variance–covariance matrix of $\hat{\theta}$

3.3 | Regression Analysis

Regression analysis is a statistical method used to figure out the relationship between variables; the regression equation describes the relationship between the dependent variable (y) and independent variable (x).

$$y = \mathrm{bx} + \mathrm{a} \tag{8}$$

The linear regression model is used as the baseline to forecast temperature in the future [13]. Linear regression, a very simple model, is used to work as the baseline in predicting the future temperature because it only models the relationship between time and target (Temperature). The prediction using this model works great as it picks up the linear trends in data. While it may be simplistic, that simple linear regression benchmark becomes a foundation to compare more sophisticated forecasting methods like ARIMA, ETS, and Holt models to assess improvements in prediction accuracy.

3.4 | Correlation Analysis

Correlation is a statistical method used to measure the degree of relationship among variables. It quantifies the relationship between two variables. The correlation coefficient is a measure of the strength and direction of that relationship, for which values range between −1 and 1. A correlation coefficient of −1 indicates a perfect negative correlation, 0 indicates no correlation, and 1 indicates a perfect positive correlation. Next, a correlational analysis is done to illuminate the relationship between extreme heat and its public health outcomes. Statistical tests, that is, Pearson correlation coefficients, are applied to evaluate the level and relevance of relationships between temperature fluctuations and health data. The Pearson correlation coefficient is a statistical measure used to assess the strength and direction of the linear relationship between two continuous variables. The Pearson correlation analysis is a common criterion to measure the correlation between variables, which is defined as [25].

$$R(i) = \frac{\sum_{k=1}^{m}\left(x_{k,i} - \overline{x_i}\right)\left(y_k - \overline{y}\right)}{\sqrt{\sum_{k=1}^{m}\left(x_{k,i} - \overline{x_i}\right)^2 \sum_{k=1}^{m}\left(y_k - \overline{y}\right)^2}} \tag{9}$$

where **m** is the number of samples, **R (i)** measures the correlation between the feature **i** and the class standard. $\boldsymbol{x_{k,i}}$, is the feature value of the feature **i** of the sample **k**. $\overline{\boldsymbol{x_i}}$ is the average value of the feature **i**. $\boldsymbol{y_k}$, is the value of the sample **k** and $\overline{\boldsymbol{y}}$ represents the class standard mean value of the sample **k**. According to the definition formula, the variation range of **R (i)** is between −1 and 1.

3.5 | Methodology for Forecast Analysis of Maximum Temperature and Diarrhea Cases

This paper employs a systematic forecasting model to examine and forecast the maximum temperature and diarrhea cases.

3.5.1 | Data Collections

The BMD data was collected from 1971 to 2020 to study the climatology of maximum temperature $\boldsymbol{T_{max}}$ trends and associated heat waves in Bangladesh. The study finds gaps in missing data in some years, filled up by the mean of the preceding and succeeding days or the long-term mean of $\boldsymbol{T_{max}}$ of the same date. A statistical technique, sampling, is used to take a few elements from many populations to make decisions. It requires the use of random selection, meaning each element in a population has an equal chance of being selected, and is most often used for data collection. This study and findings would have been developed based on collected data, which gather historical and projected climate data, including temperature, from sources such as BMD and Directorate General of Health Services (DGHS) [26–28]. The data collection period for the study has been considered from January to May 2024.

The BMD data collected from 1971 to 2020 are tabulated for the maximum and minimum temperatures recorded in Bangladesh (Table 1). This information will be used to provide an excellent basis for predicting maximum temperature changes up to 2040 and, through that, forecast climate changes affecting Bangladesh. In Bangladesh, as temperatures increase, there is a significant change in the number of diarrhea cases. Table 2 illustrates this relation between maximum temperatures and the increased

TABLE 1 | Maximum and mean temperature of Bangladesh from 1971 to 2020 (*Source:* BMD) [26].

Year	Maximum temperature (°C)	Mean temperature (°C)
1971	29.59	25.1
1972	29.6	25.6
1973	29.61	25.4
1974	29.62	25.6
1975	29.63	25.4
1976	29.64	25.6
1977	29.63	25.3
1978	29.7	25.3
1979	29.75	26.1
1980	29.08	25.8
1981	30.8	25.4
1982	30.8	25.6
1983	30.8	25.45
1984	30.8	25.6
1985	30.79	25.5
1986	30.99	25.9
1987	31.19	25.8
1988	31.39	25.9
1989	31.59	25.3
1990	31.5	25.3
1991	30.75	25.3
1992	30.7	25.3
1993	30.76	25.4
1994	30.77	25.45
1995	30.78	25.5
1996	30.79	25.55
1997	30.8	25
1998	30.8	25.6
1999	30.81	25.6
2000	30.82	25.25
2001	30.83	25.4
2002	30.84	25.35
2003	30.85	25.3
2004	30.86	25.5
2005	30.87	25.4
2006	30.88	25.6
2007	30.9	25.5
2008	31	25.7
2009	31.1	25.9
2010	31.2	26
2011	31.3	25.4
2012	31.3	25.8
2013	31.4	25.4
2014	31.4	25.6
2015	31.4	25.8
2016	31.4	26.1
2017	31.5	25.6
2018	31.5	25.8
2019	31.5	25.7
2020	31.85	26.8

TABLE 2 | Diarrhea cases in the past, source ((*Source:* DGHS), 2015–17) [27, 28].

Year	Maximum temperature (0°C)	Diarrhea cases
2000	30.82	1556
2001	30.83	1866
2002	30.84	2599
2003	30.85	2287
2004	30.86	2246
2005	30.87	2152
2006	30.88	1962
2007	30.9	2335
2008	31	2295
2009	31.1	2619
2010	31.2	2427
2011	31.3	2268
2012	31.3	2631
2013	31.4	2641
2014	31.4	2653
2015	31.4	2560
2016	31.4	2409

number of diarrhea cases. This implies that the increase in temperatures is a factor affecting diarrhea, underscoring calls for tailored public health responses to mitigate these changes. For diarrhea cases, it shows the lowest number of diarrhea cases, which was 1556 in 2000. On the other hand, in 2013, it rocketed, which was 2641 diarrhea cases in Table 2.

3.5.2 | Data Processing

The data from the health source is analyzed properly by using tools like R, Minitab, and Python to understand the status notification. This includes a cleaning stage where missing values and outliers are addressed, but also measurement unit consistency is checked. The health data is also standardized to enable robust comparison, which will enable the identification of frequent or novel associations of temperature changes with an impact on public health.

3.5.3 | Time-Series Plot of Diarrhea Cases

Time-series plots were made to illustrate how cases of diarrhea and maximum temperature were distributed with time during the study period. In addition, this method allows identifying general tendencies, variations, and possible changes in the structure of disease incidence over time. On plotting cases over time, one can see patterns behind the scenes as to dynamics over time. Moreover, the resulting visualization gave an idea of disease progression and guided further steps of analysis.

3.5.4 | Outlier Detection

To determine the extreme or anomalous observations of the maximum temperature and diarrhea case data, outlier detection techniques were used. Such procedures assist in mitigating the impact of abnormal values that could have occurred due to reporting errors or abnormal occurrences. The identification of such values is a measure to achieve the robustness and reliability of statistical and forecasting models. Therefore, the outcome of the process was cleaner datasets and enhanced model stability.

3.5.5 | Scatter Plots

Scatter plots have been used to analyze separate temporal distributions of the highest temperature and cases of diarrhea during the time frame of the study. Furthermore, the process visually represents the variability and underlying trends, and the Pearson correlation analysis measures the strength and direction of linear relationships across time. Beside this, the maximum temperature as well as the incidence of diarrhea was evaluated using a separate scatter plot.

3.5.6 | Pearson Correlation Heat Map

To visually analyze the relationship between maximum temperature and diarrhea cases, heat maps are used. Furthermore, these heat maps are used to measure the direction and magnitude of time-dependent change using Pearson correlation analysis. The intensity of color signifies the magnitude of trends, which allows the easy identification of long-term patterns of warming and trends of diseases. Furthermore, these findings suggest that there is a temperature increase or moderate rise in cases of diarrhea, which implies the application of integrated climate-health analysis. This strategy increases the level of accuracy in forecasting and supplies solid and data-driven evidence to support environmental assessment, public health monitoring, and informed policy as well as planning decisions.

3.5.7 | Trend Analysis

The trend analysis evaluated the long-term variations in the incidence of diarrhea and maximum temperature during the study period. Furthermore, the trends are crucial in assessing the development of the state of health and climate changes. The patterns identified in the analysis showed the consistent trends that shaped the modeling.

3.5.8 | Seasonal Decomposition

Decomposition methods were also used on maximum temperature and diarrhea case time series to separate out long-term components of the trend with the seasonal and irregular parts. To maximize temperature, an additive model was applied to measure underlying patterns of warming and assess seasonal stability, and based on this, dominant trend signals could be easily identified with little seasonality effect. Likewise, the data of diarrhea cases were broken down into trend, seasonal, and residual with the aim of separating the long-term dynamics of the disease and short-term fluctuations. Moreover, this combined method adds more interpretability to the temporal behavior of climatic and health data that is a solid analytical base upon which to study climate-health relationships and gauge the impact of temperature variability on the occurrence of diarrheal diseases.

3.5.9 | ACF And PACF Plots

The temporal dependence in diarrhea cases and maximum temperature was investigated using the plot of ACF and PACF. The tools determine lag structures and persistence in the time series. In addition, this kind of information is important in choosing the right time-series forecasting models. The plots helped to select the best possible parameters for the forecasting models.

3.5.10 | ADF Test for Stationary

To examine stationarity, the ADF test was used on maximum temperature and diarrhea cases. Furthermore, this statistical test establishes whether the data have unit roots which have the potential to influence the validity of the models. To be able to model time series, stationarity is a necessity. Moreover, the results of the test were used to make needed changes before making predictions.

3.5.11 | Forecasting of Maximum Temperature and Diarrhea Cases

The ETS, ARIMA, Holt, and Prophet Models were implemented to predict the maximum temperature and diarrhea cases as they represented various temporal structures and predictive accuracy. ETS represented the error, trend, and possible seasonality with the help of the state-space structure, ARIMA modeled the autocorrelation with autoregressive and moving-average terms, Holt estimated linear tendencies with the help of exponential smoothing, and Prophet flexible reflected the trends with the help of the Bayesian model. Widening 95% Monte Carlo confidence intervals were used to measure forecast uncertainty. The multi-model approach is known to have strong, plausible forecasts, allow comparison of the strength of the models, and can be utilized in credible forecasts regarding future trends of temperature and diarrheal diseases to conduct climate-health research.

3.5.12 | Regression Analysis for Diarrhea Cases vs. Maximum Temperature

The regression analysis was performed to assess the predictive relationship between maximum temperature and cases of diarrhea using an independent test dataset. The model was fitted using a linear regression model and the predicted values were compared with the observed cases to estimate the accuracy of the model. These observations, being consistent with the reference line, show that the predictive performance is reasonable, indicating that temperature variation accounts for a significant percentage of diarrheal incidence and, when assessing disease risk, temperature variation is an important climatic determinant.

3.5.13 | Evaluation of Forecasting Accuracy

The accuracy of the forecasts was measured to compare objectively the performance of the models and to determine the most valid method of doing the planning of public health. RMSE and MAE were used to measure the prediction error and average deviation to evaluate model predictions. Furthermore, the comparison among ETS, ARIMA, Holt, and Prophet models of maximum temperature and diarrhea cases was aimed at obtaining lower error, moderate error, and higher error. The findings outlined the most precise forecasting methods, which justified the choice of evidence-based models and reinforced the quality of predictions to analyze climate and health and make decisions about the health of the population.

3.5.14 | Impact of Maximum Temperature on Diarrhea

In order to determine the relationship between maximum temperature and the incidence of diarrhea in Bangladesh, we used a joint plot methodology to determine the relationship. This method produces a visual representation of the relationship between two continuous variables and at the same time shows the distribution of each variable. It can be used to easily identify patterns and possible correlations by using both marginal histograms and scatter plots. Furthermore, the approach offers an intuitive and clear structure to investigate interactions between variables and enables a strong and repeatable analysis of health-related trends of climate.

In this study, a systematic methodology for forecast analysis of maximum temperature and diarrhea cases is implemented. This is important because those predictions feed into the development of strategies for dealing with extreme heat events. The insights obtained from this will be useful for the policymakers and public health officials to formulate targeted interventions, thereby improving the adaptive capacity of vulnerable populations in Bangladesh which contributes towards building resilience against climate change with time.

4 | Results and Discussion

The results of the study along with the discussion are presented in this section, including variations analysis of maximum and mean temperatures between the years 1971 to 2020, time series plots for maximum temperature, as well as forecasting using the ETS, Holt, ARIMA, and Prophet Models. The results are compared with the existing approach for a better understanding of the accuracy of forecasting and its public health implications in Bangladesh.

4.1 | Model Training and Validation

The dataset was examined using the R and Python programming languages. The whole experimentation was carried out on a system operating on Windows 10 with Python 3.10, Intel Core i7 (2.4 GHz), 16 GB RAM and 1 TB storage. All the models were developed in VS code and Kaggle using machine and deep learning libraries. Seventy percent of the dataset was used for training, while the remaining 30% was used for testing purposes (i.e., the years 2012–2016). The test dataset was applied to evaluate their performance.

4.2 | Maximum and Mean Temperatures (1971–2020)

The analysis shows variation in maximum and mean temperatures in Bangladesh over the period from 1971 to 2020. In Figure 1, the time series plot exhibits maximum and mean temperatures from 1971 to 2020. In 2020, the maximum temperature was 31.85°C which was the highest temperature from 1971 to 2020. On the other hand, in 1971, the maximum temperature was

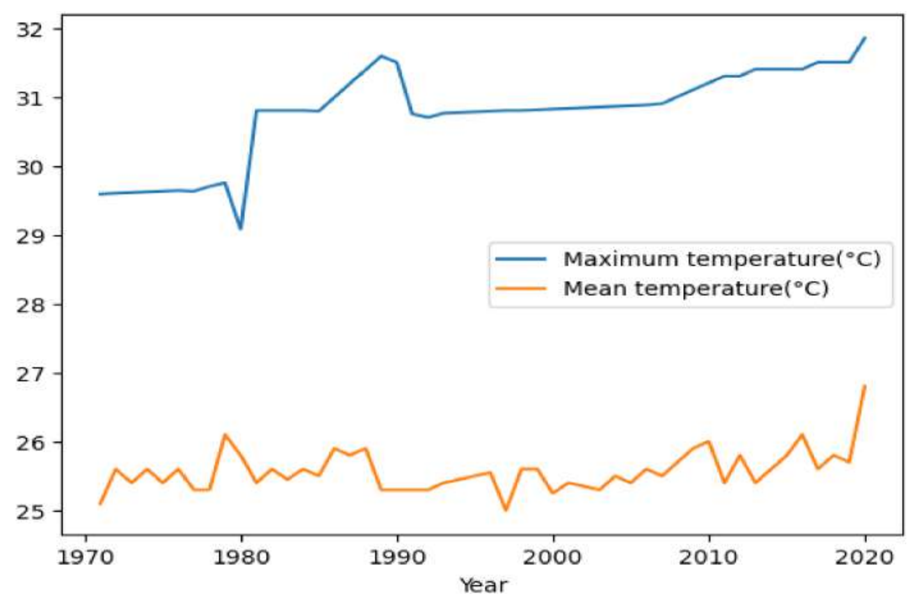

FIGURE 1 | Variation in maximum and mean temperature (1971–2020).

29.59˚C which was the lowest maximum temperature from 1971 to 2020. The mean temperature was 26.8˚C in 2020 which was the highest mean temperature from 1971 to 2020. However, in 1971, the mean temperature was 25.1˚C which was the 2nd lowest mean temperature from 1971 to 2020.

4.3 | Time Series Plot and Trend Analysis Plot for Maximum Temperature (˚C)

The study reveals important trends in the climate of the country through time series data for maximum temperatures. In Figure 2, the time series plot exhibits the maximum temperature from 1971 to 2020. In 2020, the maximum temperature was 31.85˚C which was the highest temperature from 1971 to 2020. On the other hand, in 1971, the maximum temperature was 29.59˚C which was the lowest maximum temperature from 1971 to 2020. For the years 2000 through 2016, this study analyzes temperatures to catch outliers and check for correlations. In Figure 3, the data set including outliers was cleaned by identifying them with the Interquartile Range (IQR) approach and removing them, and the boxplots of the two datasets are quite similar, which suggests outliers made only little difference to show the data is distributed. There exists a strong relationship ($r = 0.955$) in the scatter plot between rising temperatures and passage of time in Figure 4. The graph in Figure 5 also proves the relationship, since year and temperature often have high correlation values (0.95–1.00). It is evident from these studies that temperatures keep rising during the observable time.

As shown in Figure 6, the trend analysis for maximum temperature observed a much higher trend of temperature over the years indicating a clear upward trend in temperatures of Bangladesh. The paper uses RMSE, MAPE (Mean Absolute Percentage Error), MAD (Mean Absolute Deviation), and MSD (Mean Squared Deviation) to evaluate the accuracy of its time series model for maximum temperature trajectory. This is because it penalizes big errors more than small ones; RMSE is used to find models with less variation, especially when large differences are remarkable. MAPE enables one to quickly see how wrong the predictions can be by expressing the percentage error. This examined three metrics of accuracy in Figure 6, one being MAPE (0.945801), followed by MAD of 0.28928 as well as MSD of 0.160156. RMSE is the square root of MSD, and when MSD = 0.160156, RMSE is estimated at 0.4002. This implies that the model tends to make predictions close to reality, and its predictions are usually uniformly correct. MAPE below 1% points to strong predictive precision. According to the decomposition in Figure 7, the maximum temperature (2000–2016) shows a positive trend that indicates continuous warming, distinct seasonal changes with yearly peaks and valleys, and only minor changes, meaning the temperature variations are repetitive. This relationship agrees with what was seen in the earlier strong correlation ($r = 0.955$), providing more evidence of global warming.

The retained maximum temperature series has ACF and PACF plots shown in Figure 8. The ACF exhibits a decreasing pattern and noticeable peaks at lags 1 and 2 which suggest that the series is auto correlated for the short term. It can be seen from the PACF plot that there is a marked reduction after lag 2. ADF test statistics are −2.9573 and the p value is 0.0391, so this is significant at the 5% level in Table 3. Thus, the null hypothesis of a unit root is dismissed, showing that the series is stationary after differencing. The combination of the plots as well as statistical data indicates that the ARIMA model is appropriate, and that applying differencing effectively removes trajectory and non-stationary.

4.4 | Forecasting of Maximum Temperature Using ETS, ARIMA, Holt and Prophet Models

In Figures 9–12, it has been observed that, to forecast maximum temperatures from 2017 to 2040, four time series models such as ETS, ARIMA, Holt and Prophet Model were deployed. Each model was evolved with information from past temperatures (2000–2011) and evaluated using a distinct 2012–2016 test dataset. Every model exhibits strong accuracy in forecasting compared to what was observed in the data. According to a consensus of the models, there was a pellucid rising trajectory in the

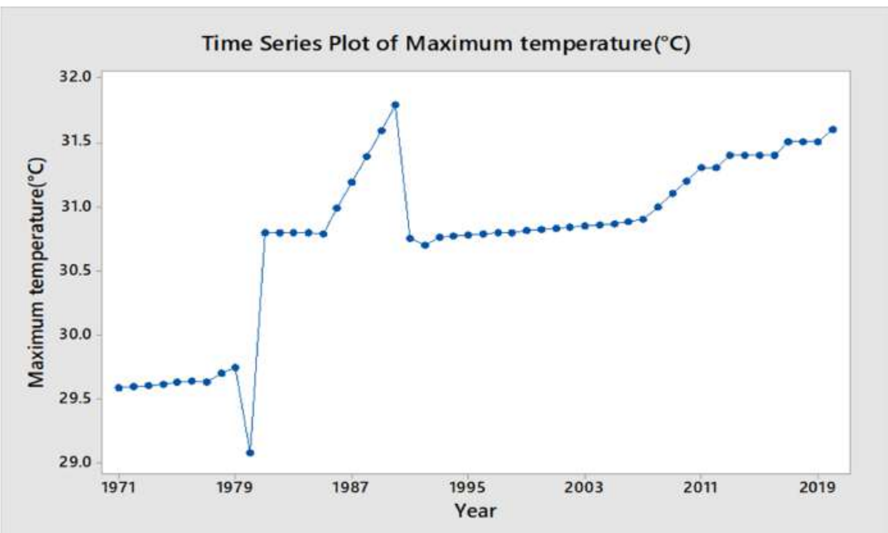

FIGURE 2 | Time series plot for maximum temperature (˚C).

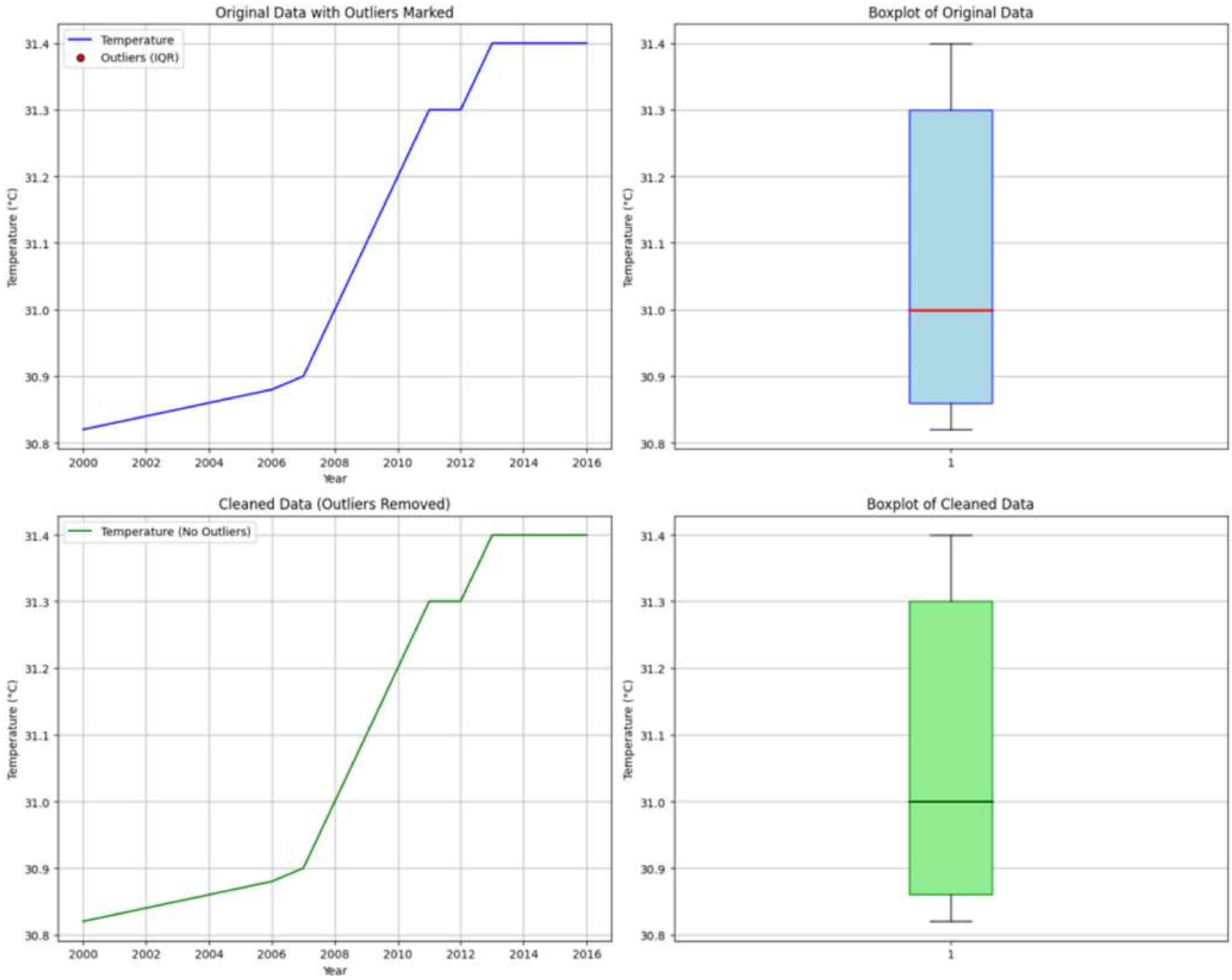

FIGURE 3 | Outliers of maximum temperature.

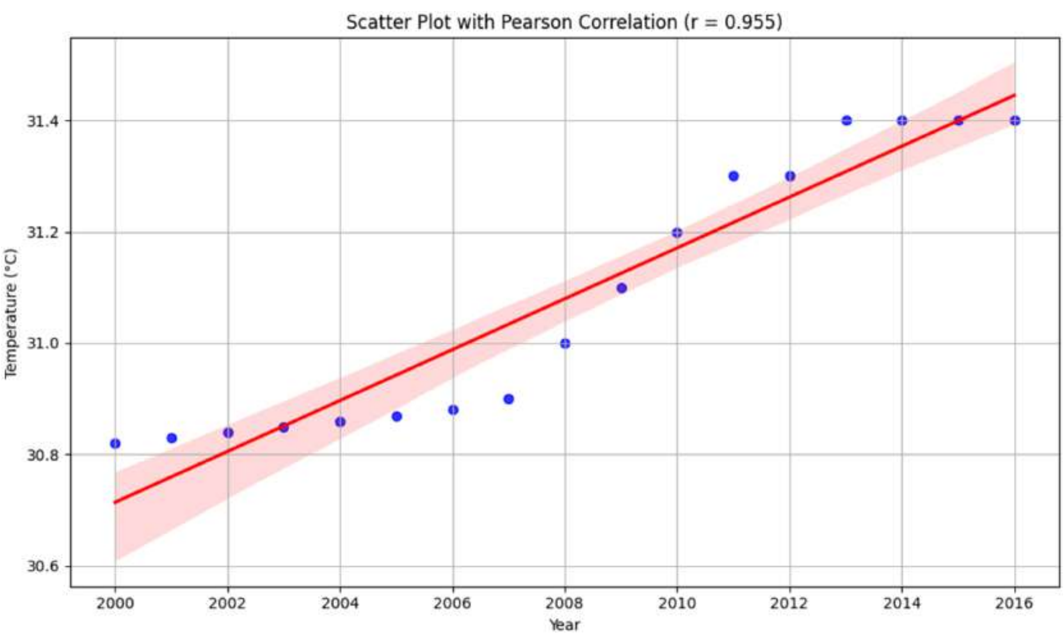

FIGURE 4 | Scatter Plot with Pearson correlation of maximum temperature (˚C).

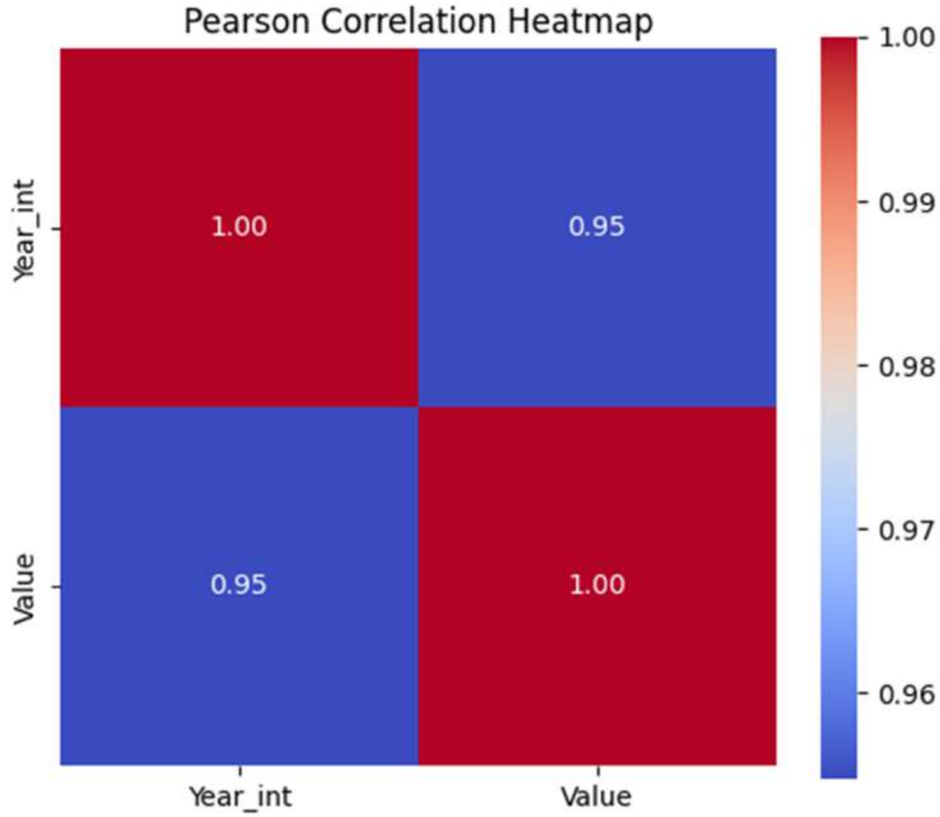

FIGURE 5 | Pearson correlation heat map of maximum temperature (˚C).

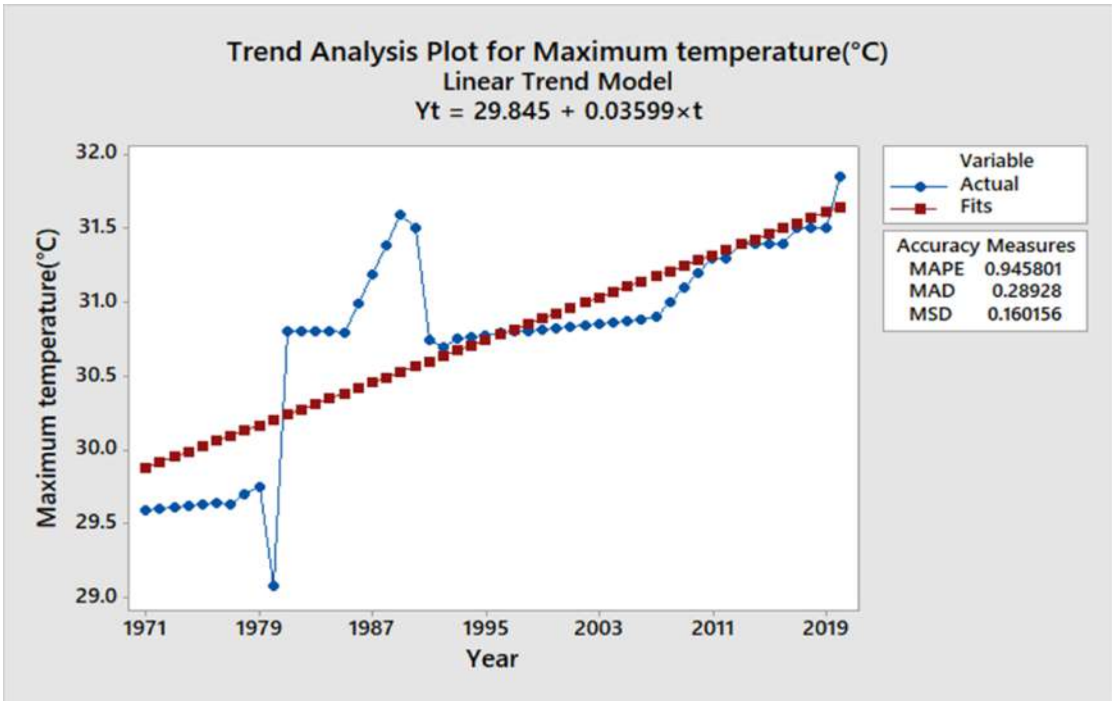

FIGURE 6 | Trend analysis plot for maximum temperature (˚C).

maximum temperatures. The ARIMA model forecasted a modest increase in maximum temperature from approximately 31.39˚C in 2017 to 32.15˚C by 2040, the widening Monte Carlo 95% confidence interval, ranging from 31.34˚C to 33.99˚C in Table 4. In comparison, the ETS experienced a temperature rise, where the maximum temperature moved from 31.41˚C to 33.71˚C in the same time span. Because the model's confidence bands are narrow, its forecast is very stable. Such as the other models, Holt's

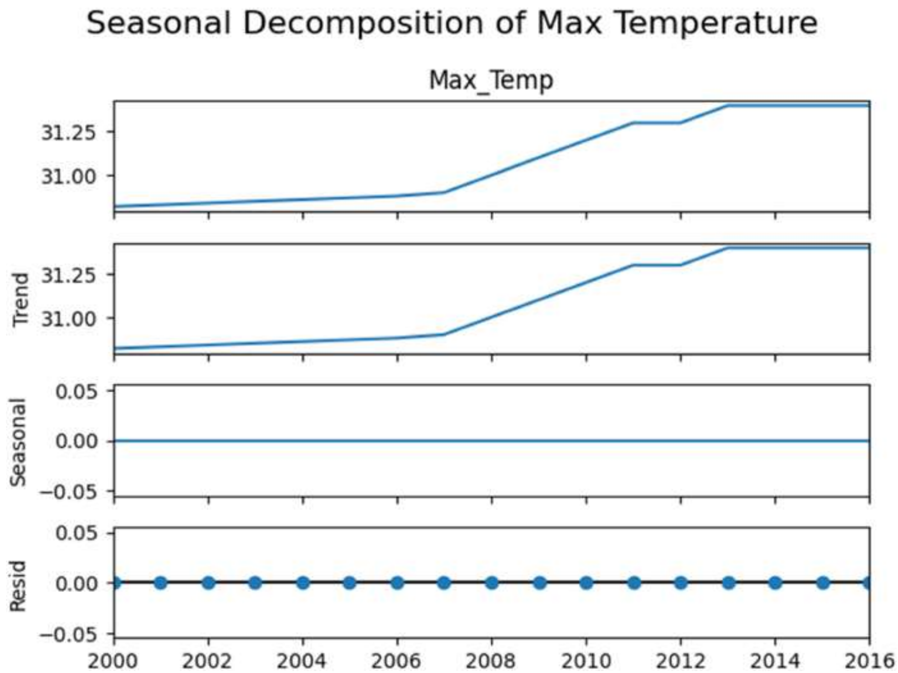

FIGURE 7 | Seasonal decomposition of maximum temperature (˚C).

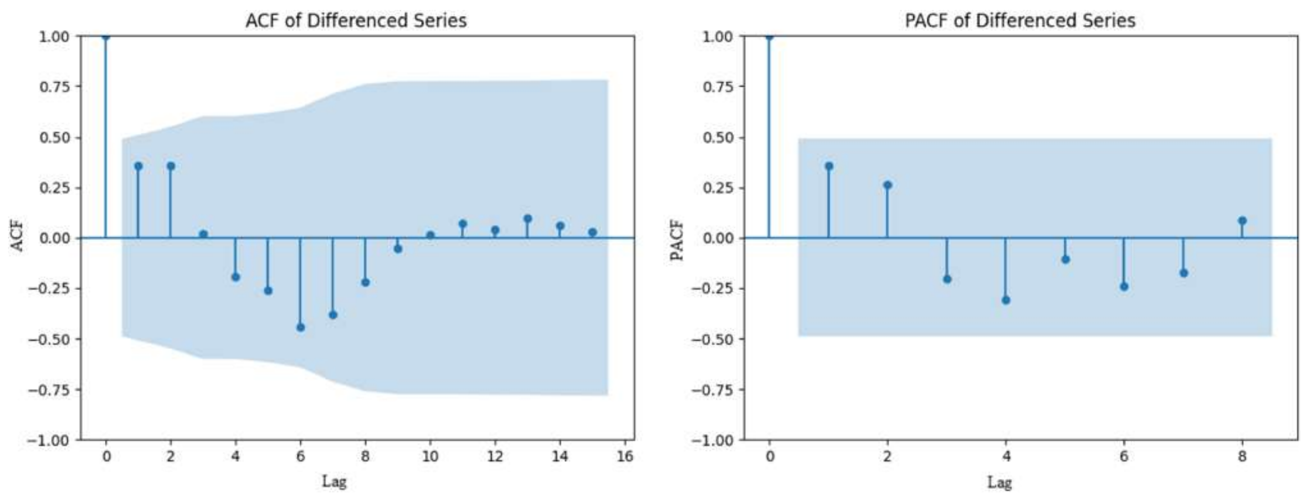

FIGURE 8 | ACF and PACF plot for maximum temperature (˚C).

TABLE 3 | ADF test for maximum temperature (˚C).

ADF Statistic: −2.9573
p value: 0.0391
Lags Used: 6
Observations Used: 9
Critical Value (1%): −4.4731
Critical Value (5%): −3.2899
Critical Value (10%): −2.7724
Conclusion: The differenced time series is stationary.

model showed a gradual increase in temperature that is predicted to reach about 33.71˚C by 2040 with reasonable confidence. In addition, the Prophet data estimates that by 2040, temperatures will reach 34.3˚C. Furthermore, this model showed wider confidence intervals. All models showed useful predictions regarding future changes in temperature. As a result, both the ability to forecast correctly and the way the model describes climate trends matter when choosing a model for climate-based decisions.

According to the provided Table 5, it represents that the forecasting performance of four time series models such as ARIMA, ETS, Holt, and Prophet using RMSE and MAE to assess predictive accuracy. ARIMA achieved the lowest RMSE (0.18) and MAE (0.16), indicating superior accuracy and minimal deviation from actual temperature values. In contrast, ETS, Holt, and Prophet models all recorded higher and identical error values (RMSE: 0.25, MAE: 0.22), suggesting lower predictive accuracy. However, those models may capture general trajectory, ARIMA's lower error rates highlight its robustness. As a result, ARIMA

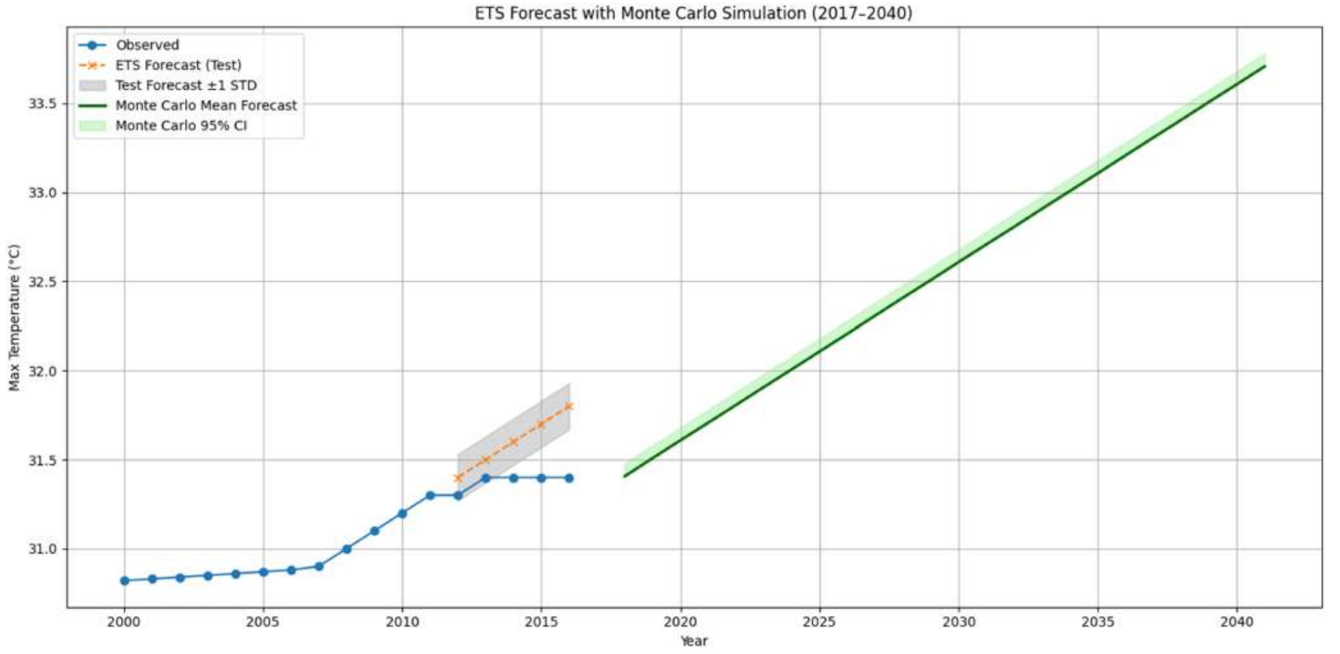

FIGURE 9 | ETS Model forecasting analysis plot for maximum temperature (°C).

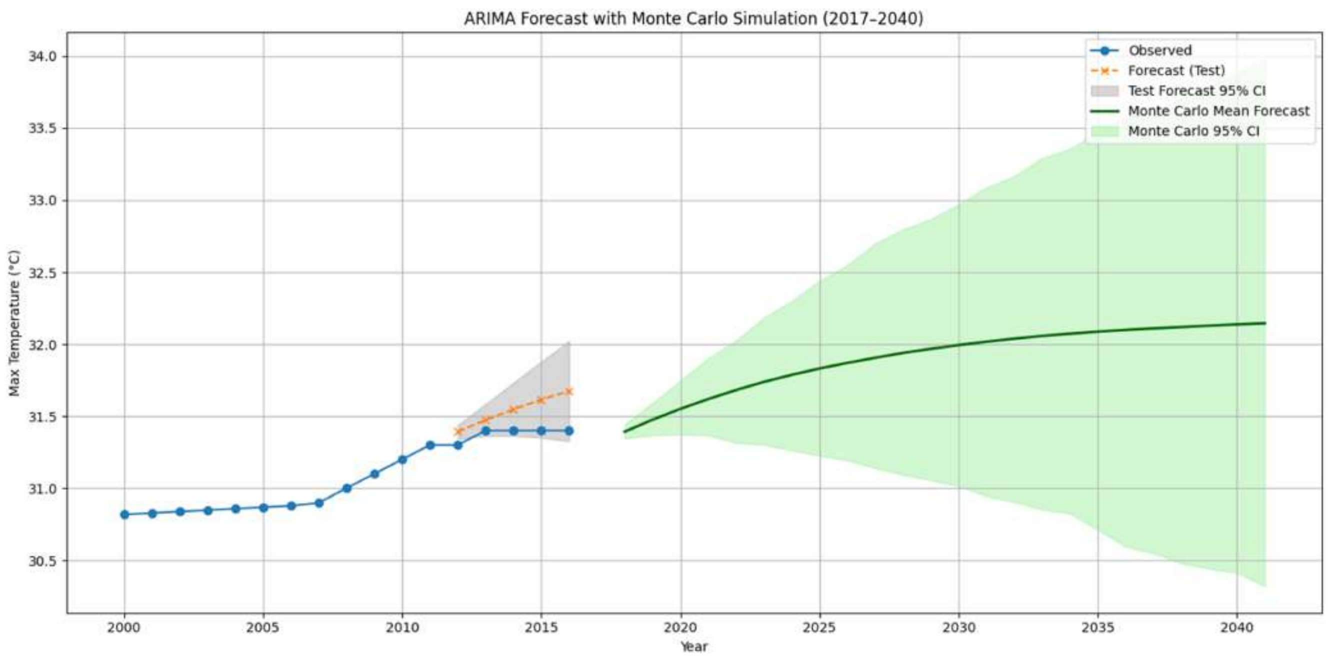

FIGURE 10 | ARIMA model forecasting analysis plot for maximum temperature (°C).

is statistically the most reliable model for forecasting maximum temperature.

4.5 | Diarrhea Cases Time Series Plot and Trend Analysis

The time series plot for cases is shown in Figure 13 that illustrates the fewest number of diarrhea cases comprising just 1556 in 2000. However, in 2013 it rocketed, reaching 2641 cases of diarrhea. So, the time series plot exhibits a clear upward trend regarding the diarrhea cases of Bangladesh during 2000 to 2013. Diarrhea cases were examined regarding changing patterns using various statistical approaches. The first step in data visualization in Figure 14 used the IQR technique to remove outliers and maintain the quality of the data. There was a strong positive relationship between the year and number of cases found in the data ($r = 0.68$, $p < 0.05$) in Figure 15. The correlation heatmap clearly verified the order of events and showed that certain variables were strongly associated (0.68–1.00) in Figure 16. From the Figure 17 shows the

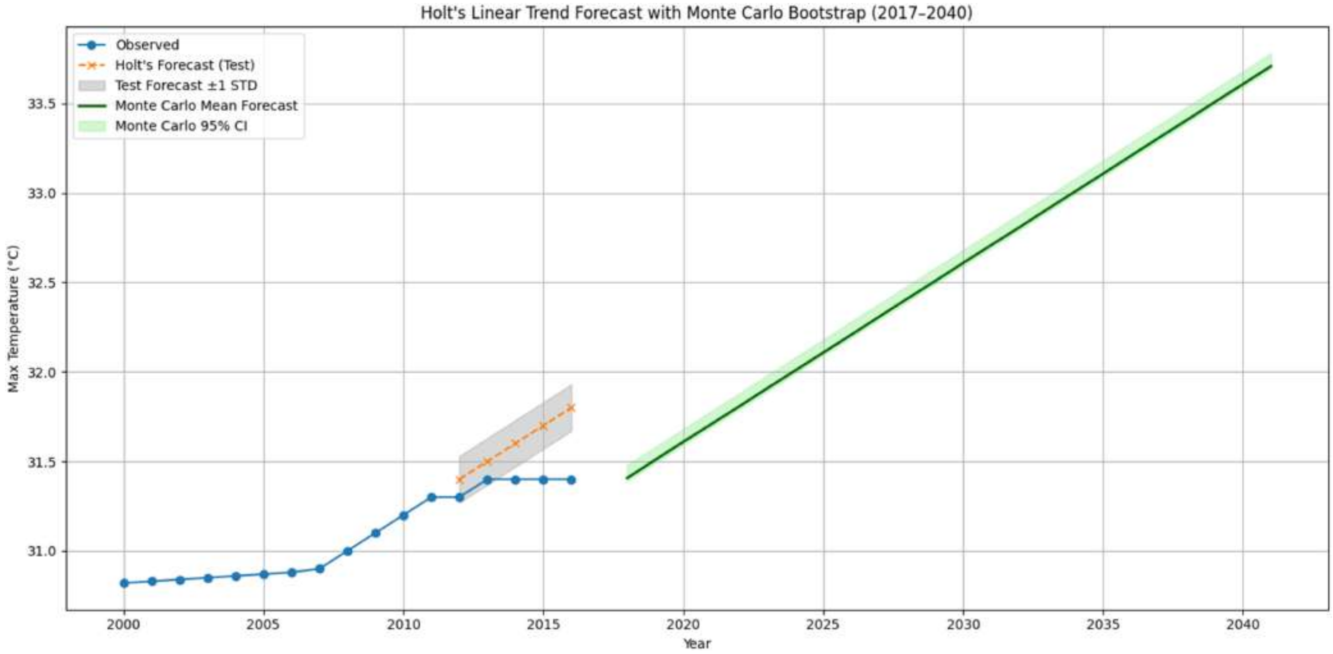

FIGURE 11 | Forecasts for maximum temperature (°C) Holt model.

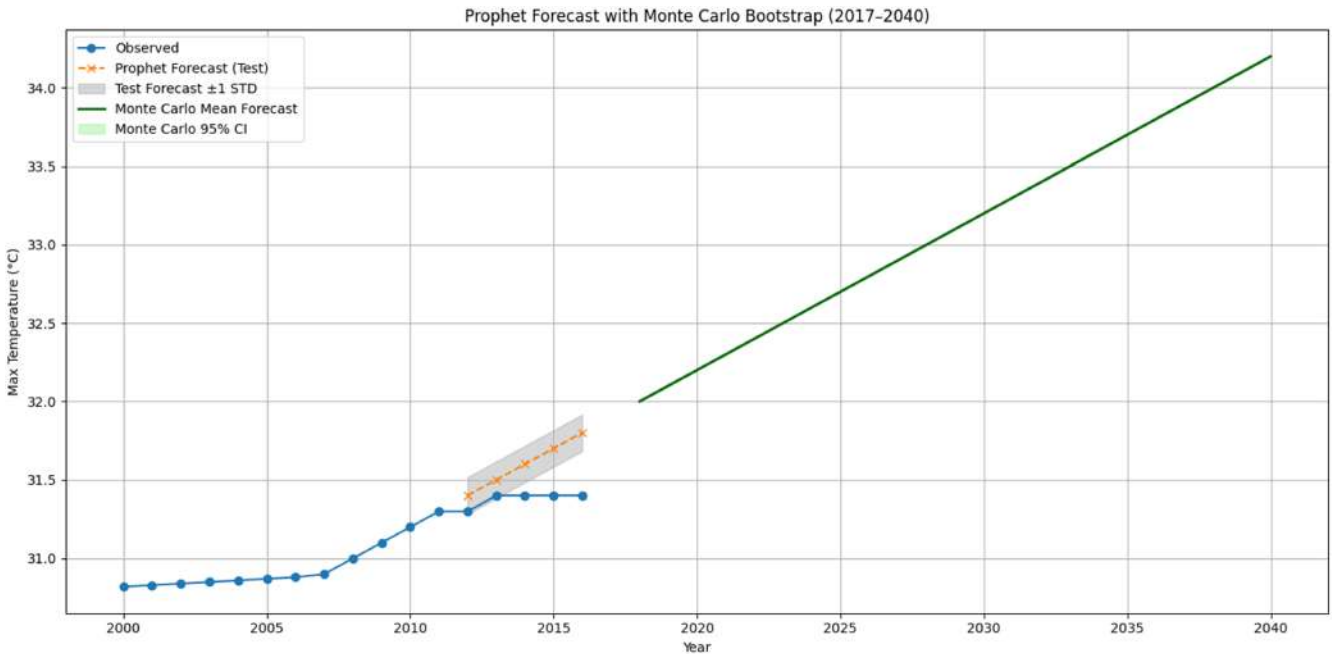

FIGURE 12 | Prophet model forecasting analysis plot for maximum temperature (°C).

trend analysis plot of the diarrhea cases pattern from 2000 to 2013. Each year presents a high-inclining curve, reflecting the continual rise in the number of cases each passing year. RMSE, MAPE, MAD and MSD are used by the study to measure the accuracy of its model for diarrhea cases. The approximately RMSE value in Figure 17 is 226.5 and MAPE is 7.9, MAD is 169.5 and MSD is 51,324.2. A MAD of 169.5 explains that on average, predictions are not the same as the real values by this amount, which is a comfortable fit. Thus, the plot exhibits that while the actual number of diarrhea cases fluctuates significantly, the fitted trend line suggests a steady increase in the number of cases over time. Hence, the time series plot for diarrhea cases of Bangladesh during 2000 to 2012 clearly indicates an increasing trend. When the data was analyzed with a seasonal decomposition of diarrhea cases in Figure 18, three key elements were found: rising cases reflected by an upward trend component, clear evidence of seasonal cycles in diseases, and the residuals were quite small, confirming that the model fitted the data well. This study proves that diarrhea occurs more in some seasons and with a general growth over the years.

TABLE 4 | Comparison of maximum temperature (°C) forecasting using ETS, ARIMA, Holt, and prophet models.

Year	ETS point forecast	ARIMA point forecast	Holt point forecast	Prophet point forecast
2017	31.41	31.39	31.41	32
2018	31.51	31.47	31.51	32.1
2019	31.61	31.55	31.61	32.2
2020	31.71	31.61	31.71	32.3
2021	31.81	31.67	31.81	32.4
2022	31.91	31.73	31.91	32.5
2023	32.01	31.77	32.01	32.6
2024	32.11	31.82	32.11	32.7
2025	32.21	31.85	32.21	32.8
2026	32.31	31.89	32.31	32.9
2027	32.41	31.92	32.41	33
2028	32.51	31.95	32.51	33.1
2029	32.61	31.97	32.61	33.2
2030	32.71	32	32.71	33.3
2031	32.81	32.02	32.81	33.4
2032	32.91	32.04	32.91	33.5
2033	33.01	32.06	33.01	33.6
2034	33.11	32.08	33.11	33.7
2035	33.21	32.09	33.21	33.8
2036	33.31	32.11	33.31	33.9
2037	33.41	32.12	33.41	34
2038	33.51	32.13	33.51	34.1
2039	33.61	32.14	33.61	34.2
2040	33.71	32.15	33.71	34.3

TABLE 5 | Comparison of maximum temperature error function of ETS, ARIMA, Holt, and prophet models.

Model	RMSE	MAE
Prophet	0.25	0.22
ETS	0.25	0.22
ARIMA	0.18	0.16
Holt	0.25	0.22

The ACF and PACF plot for diarrhea cases has shown in Figure 19. The ACF plot indicates that there is little or no correlation after lag 5 and the series is not highly dependent over time. The PACF plot exhibits a noticeable increase at lag 5, possibly indicating that the series is autoregressive. After running the ADF test, it found a score of -4.76 and a p value less than 0.001 in Table 6. It strongly supports the alternative hypothesis, which means that the transformed series is stationary. Such results indicate that ARIMA modeling can be used for this data since the system is stationary and patterns in diarrhea cases can thus be analyzed and predicted.

4.6 | Forecasting of Diarrhea Cases Using Prophet, Holt, ARIMA, and ETS Models

From the provided graph, it has been observed that four time series forecasting models such as ARIMA, ETS, Holt, and Prophet model were applied to forecast diarrhea cases and maximum temperature trends from 2017 to 2040 (Figures 20–23). The comparison of diarrhea cases forecasting of Prophet, Holt, ARIMA, and ETS models is delineated in Table 7. Each model demonstrates different methods of forecasting and different performance results. The ARIMA forecast exhibits an increase in diarrhea cases each year, going up from about 2620 cases annually in 2017 to 4045 cases in 2040. The Monte Carlo 95% confidence interval stretches, meaning there is more uncertainty. In addition, there is a MAE of 155.94 and RMSE of 192.30 for the ARIMA model's forecasts on the 2012–2016 test set, which means the forecasts are accurate and have low variation. According to the ETS model, the number of diarrhea cases was predicted to grow constantly, commencing at approximately 2561 in 2017 and terminating at around 3786 in 2040. According to the Holt model, diarrhea cases are predicted to rise slightly from 2017 to 2040, roughly from 2198 to 2282 cases. On the flip side, the Prophet model suggests that the number of cases was about 2788 in 2017; however, it will rise to an estimated 3841 by 2040.

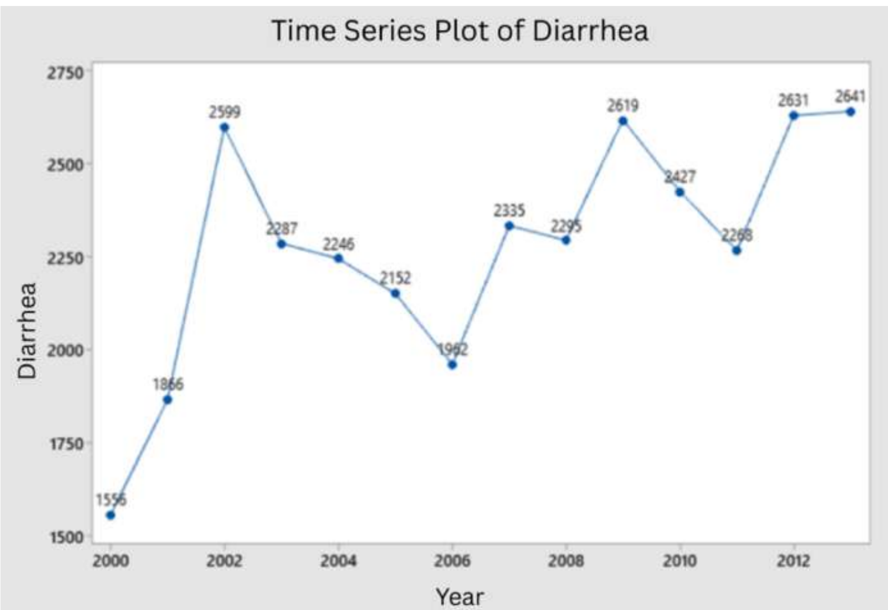

FIGURE 13 | Time series plot for diarrhea cases in the past.

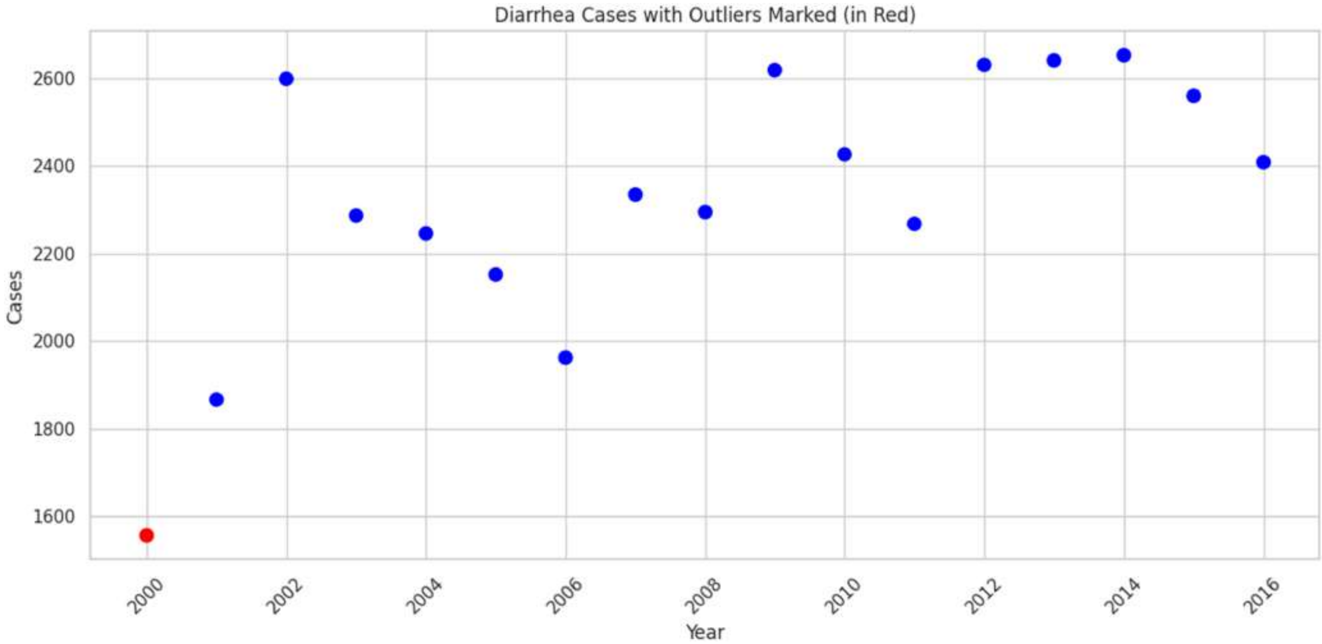

FIGURE 14 | Outliers of diarrhea cases.

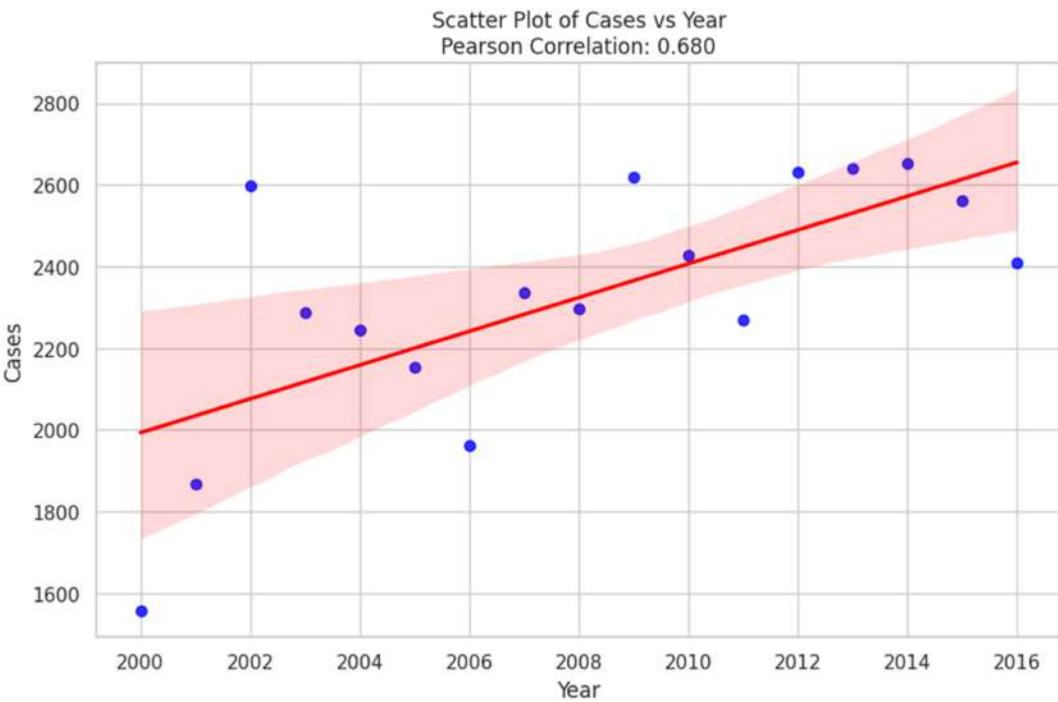

FIGURE 15 | Scatter plot with Pearson correlation of diarrhea cases.

The comparison of diarrhea cases error function of ETS, ARIMA, Holt and Prophet Models demonstrated in Table 8. Prophet model achieved the lowest MAE (124.69) and RMSE (151.62), indicating the highest forecast accuracy. ETS followed closely with a slightly higher MAE (125.15) and a larger RMSE (176.21). Furthermore, Holt's model illustrated moderate performance with MAE as well as RMSE values of 408.22 and 402.38, respectively. On the flipside, ARIMA model demonstrated the lowest score, having MAE of 155.94 and RMSE of 192.30 which highlights its lack of suitability in this case. In forecasting diarrhea cases, Prophet showed the highest accuracy and ARIMA and ETS followed with moderate accuracy.

4.7 | Impact of Maximum Temperature on Diarrhea in Bangladesh

In Bangladesh, as temperatures increase there is a considerable change in the number of diarrhea cases. The joint plot of the relationship between cases of diarrhea and maximum

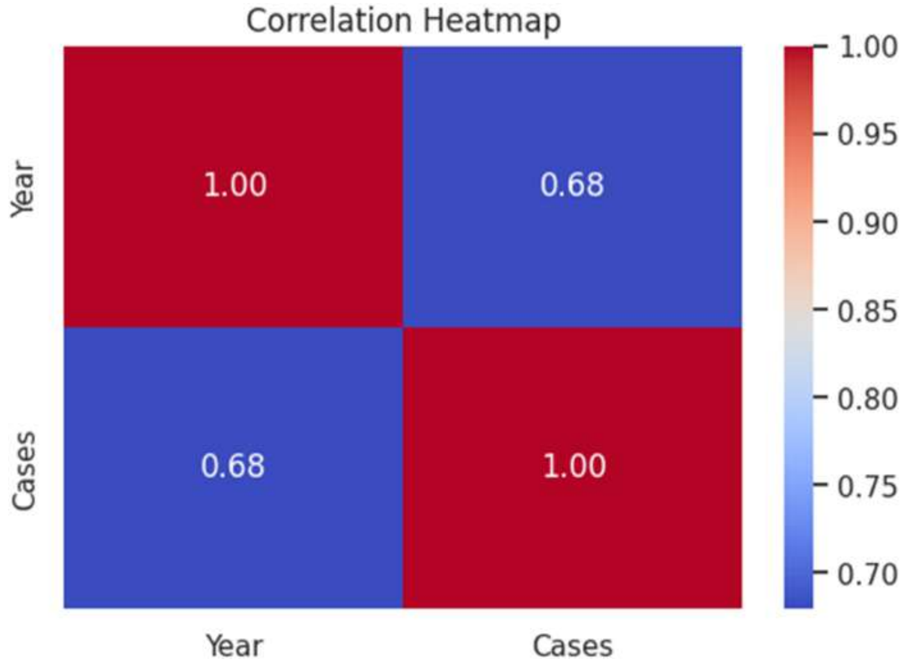

FIGURE 16 | Pearson correlation heat map of diarrhea cases.

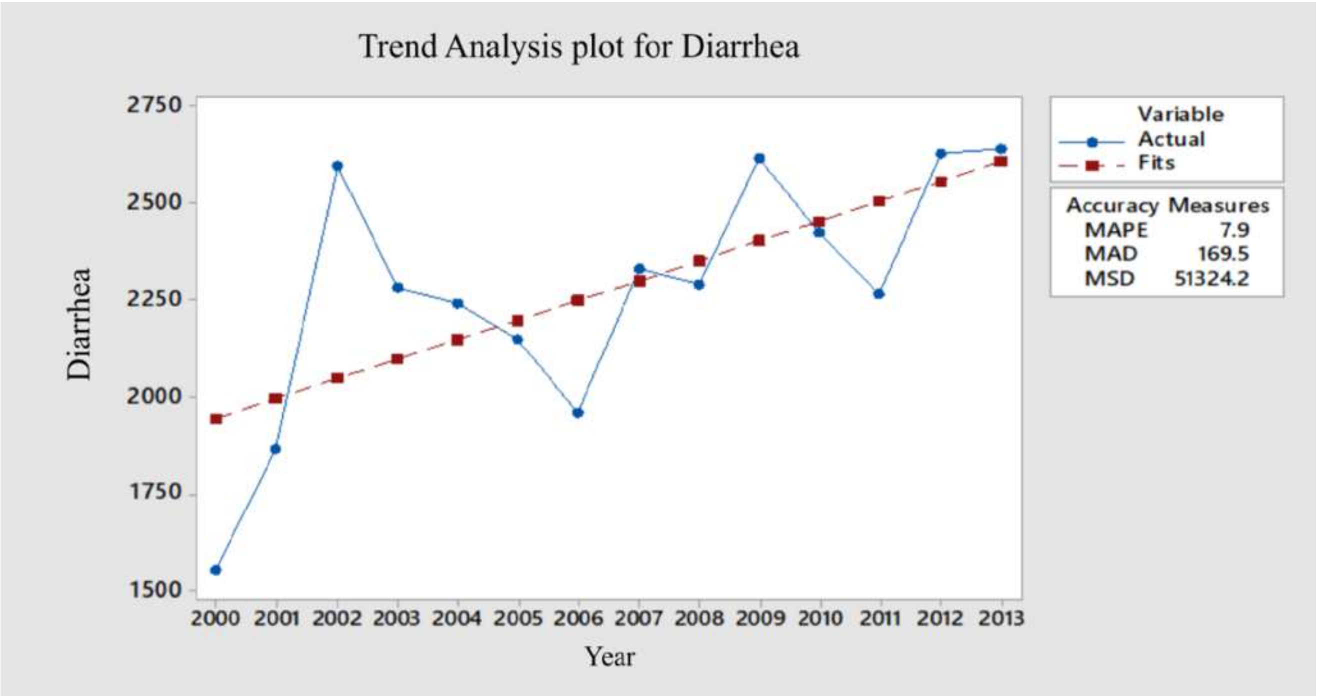

FIGURE 17 | Trend analysis plot for diarrhea cases in the past.

temperature (in ˚C) is shown in Figure 24. The central panel tells us the relationship, with side histograms showing the distributions of each variable. The plot shows that by increasing the temperature, diarrhea increases as well and this suggests a prospective positive relation between these two variables. When the maximum temperature is rising, cases of diarrhea go up. This can be seen from the upward trend of the scatter plot, suggesting a positive correlation. The histogram is near-uniform along the *x*-axis which depicts a wide range of maximum temperatures between 30.8˚C and 31.4˚C observed over our data set. The histogram on the y-axis shows that the cases of diarrhea are between 1600 and about 2800, demonstrating a broad distribution. In general, the joint plot provides a clear visual representation of the positive correlation between maximum temperature and diarrhea cases. Overall, this visual analysis exhibits patterns in the data without exaggerating causality, which gives a translucent and accurate summary of the observed correlation between temperatures and diarrhea.

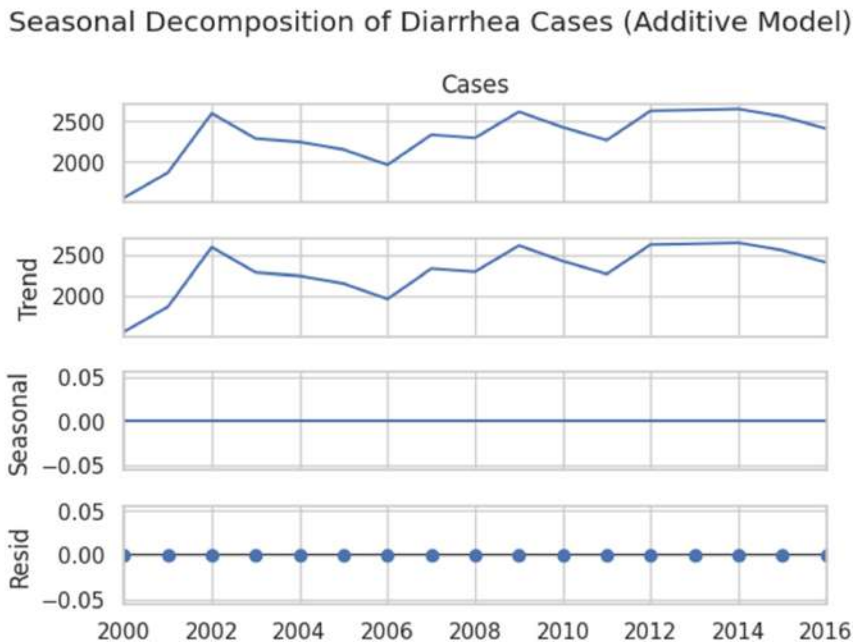

FIGURE 18 | Seasonal decomposition of diarrhea cases.

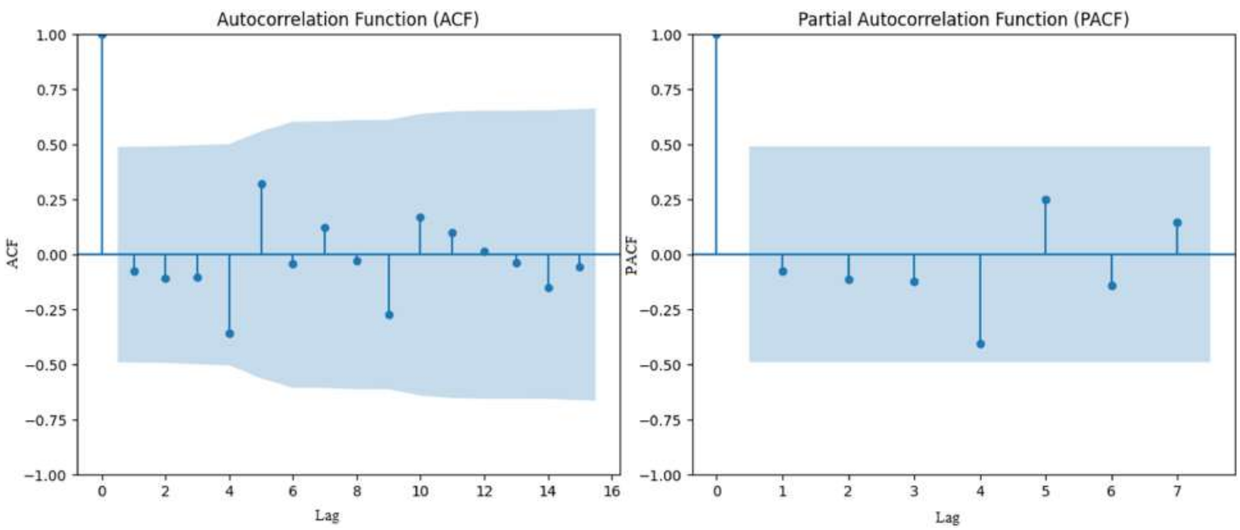

FIGURE 19 | ACF and PACF plots for diarrhea cases.

TABLE 6 | ADF test for diarrhea cases.

ADF Statistic: −4.7601404604325
p value: 6.478366458296168e-05
Critical Values:
1%: −4.223238279489106
5%: −3.189368925619835
10%: −2.729839421487603
The log-transformed and differenced series is stationary.

4.8 | Regression Analysis for Diarrhea Cases vs. Maximum Temperature in Bangladesh Perspective

As Figure 25 shows, a Pearson correlation coefficient of 0.60 indicates a moderately positive linear relationship exists between the two variables. The correlation coefficient (r) can take a value ranging between −1 and 1. A significant correlation which does not occur randomly will use the p value to determine statistical significance of the correlation coefficient. The p value ($p = 0.024$) exhibits statistical significance (less than 0.05) which confirms a significant correlation between diarrhea cases and maximum temperature.

The coefficient of determination (R^2) revealed maximum temperature alterations that explain 35.90% of all variations in reported diarrhea cases. A predictive value of 35.90% counts as acceptable in public health research even though higher R^2 values are standard in practice. The model demonstrates a statistically significant relationship ($p = 0.024$) at a low but positive correlation ($r = 0.60$) between maximum temperature and diarrhea incidence although the R^2 value remains moderate.

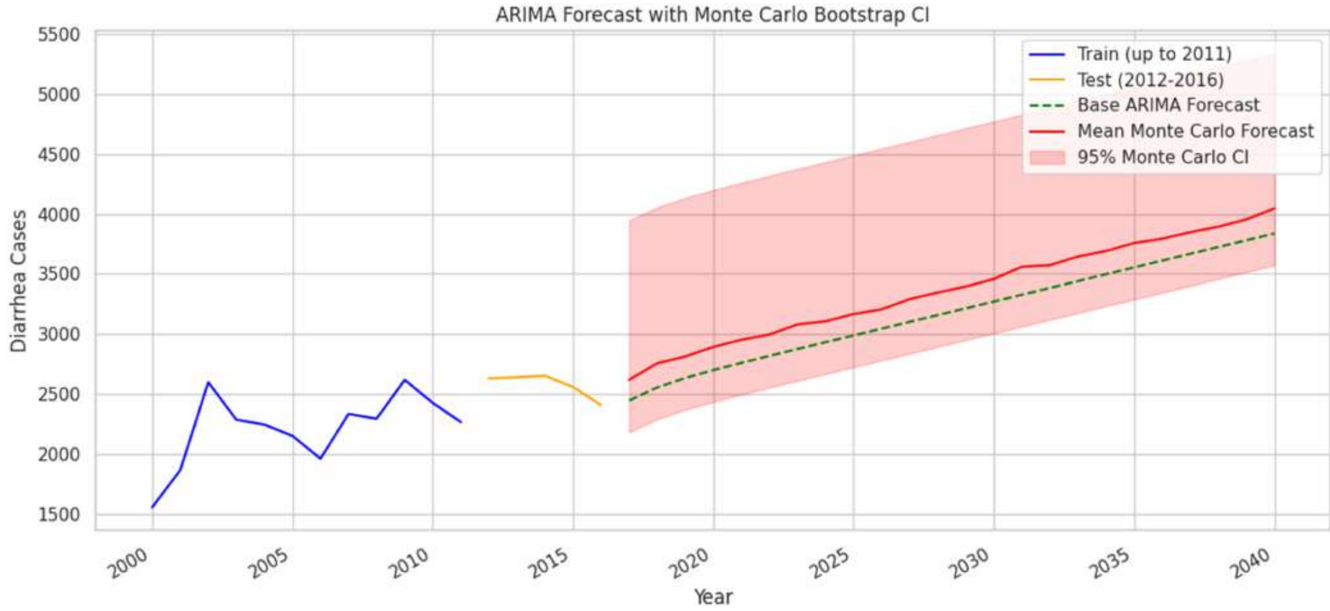

FIGURE 20 | Diarrhea cases forecasts using ARIMA model.

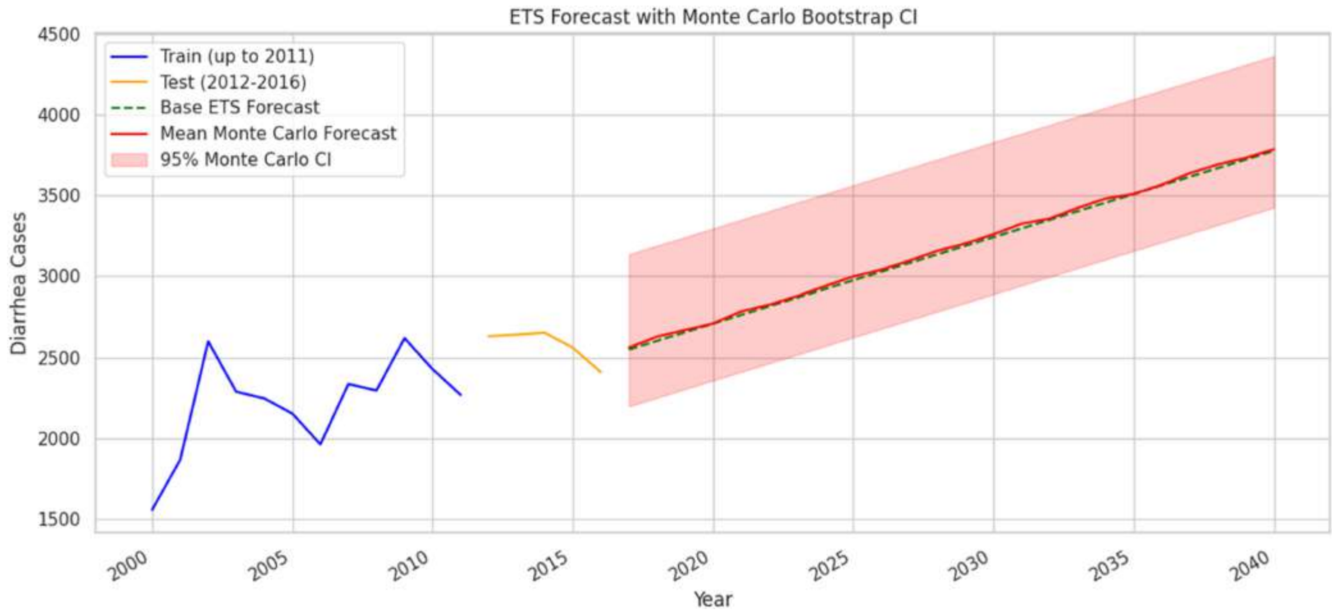

FIGURE 21 | Diarrhea cases forecasts using ETS model.

Pearson correlation coefficient of 0.60 represents a moderate positive linear relationship between two variables; it tends to move together in the same direction positively but not perfectly. The fitted equation for the linear model that describes the relationship between Y and X is as follows:

$$Y = 26167 + 917.2X$$

From Figure 25, Regression model indicates that if the model fits the data well, this equation can be used to predict diarrhea cases for a value of maximum temperature, or finds the settings for Maximum temperature that correspond to a desired value or range of values for Diarrhea cases.

In Figure 26, the relationship between actual and predicted cases of diarrhea in the test set is shown with the scatter plot. The vast majority of data points are near the 45-degree line, which is proper for the accuracy of prediction. The pattern as a whole demonstrates a strong association and reliable predictions for short-term diarrhea cases, proving the model helps in diarrhea prediction.

A strong relationship between higher temperatures and an increase in diarrhea cases has been established by the regression model shown in Tables 8 and 9. It exhibits that with a 1°C rise in the maximum temperature, the number of diarrhea cases increases by about 816.3°C, all other things being constant.

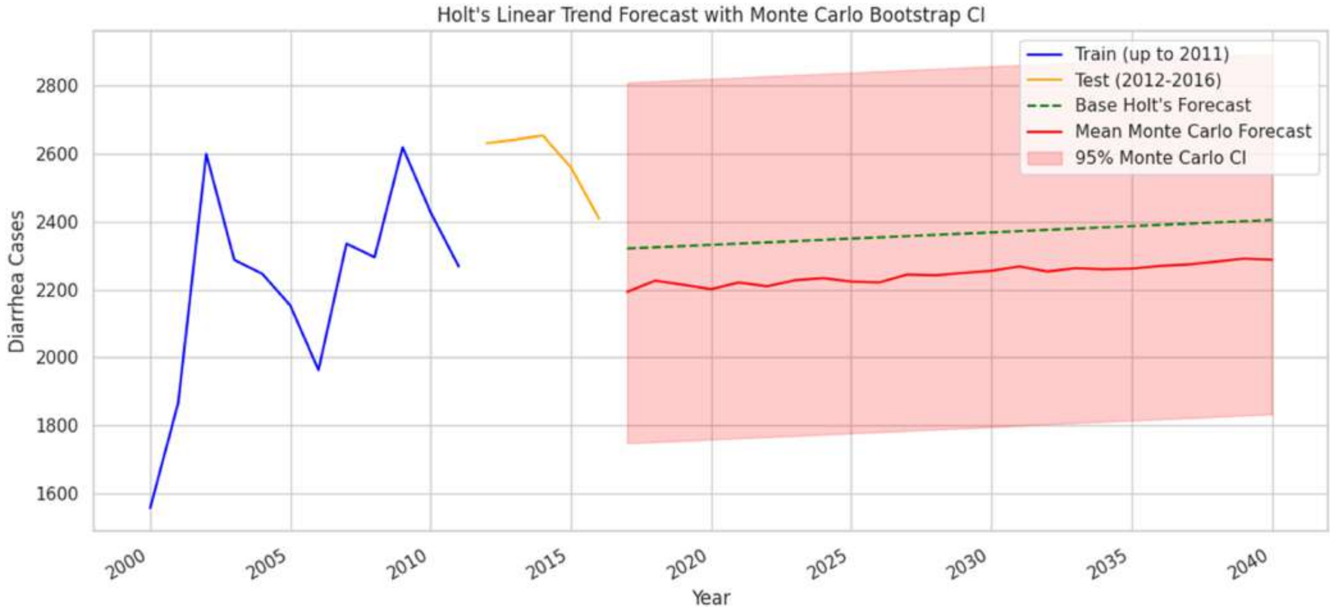

FIGURE 22 | Diarrhea cases forecasts using Model Holt model.

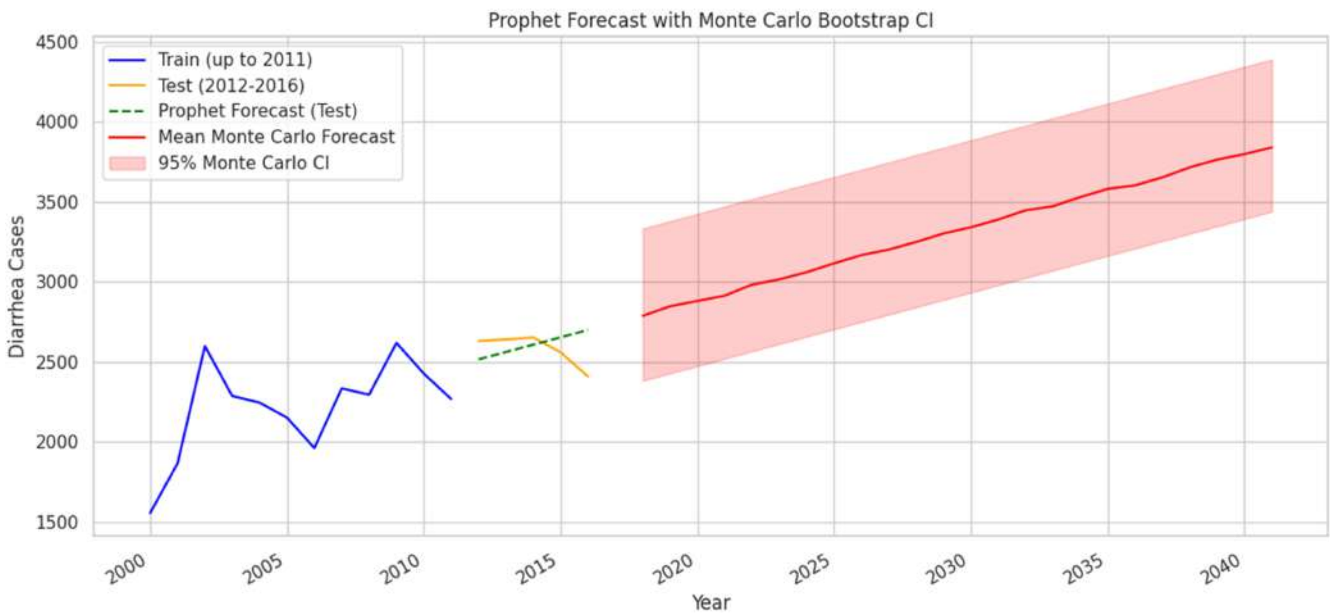

FIGURE 23 | Diarrhea cases forecasts using prophet model.

The 5% significance level ($p < 0.05$) applies to both coefficients. Out of the Train dataset, the model has an R-squared value of 0.381, which explains 38.1% of the variation and a residual standard error of 268.2. The model's R-squared on training data is about 0.38—meaning ∼38% of variance explained. On the test set, R-squared improved to 0.68, which is quite solid. MAE (∼135) and RMSE (∼155) show the average prediction errors in Table 10. The generated plot helps to visualize how well predicted values align with actual values (points close to the red dashed $y = x$ line indicate good predictions). There is a strong and positive connection between high temperature and diarrhea cases, as shown by the regression model. With improved R-squared (0.68) on the test dataset and reasonably low MAE and RMSE values, the model shows strong short-term predictive capability.

4.9 | Discussion

In this discussion part, a brief comparison is made between this work and the recent state-of-the-art methodologies of maximum temperature and average temperature in Bangladesh. Furthermore, the study examines how temperature changes are correlated with cases of diarrhea and underscores that a reliable model must be used to prepare for this health impact of climate change in Bangladesh.

TABLE 7 | Comparison of diarrhea cases forecasting of prophet, Holt, ARIMA, and ETS models.

Year	ARIMA	ETS	Holt	Prophet
2017	2620	2561	2198	2788
2018	2757	2628	2199	2848
2019	2814	2669	2220	2881
2020	2893	2709	2202	2914
2021	2953	2784	2217	2983
2022	2997	2825	2243	3016
2023	3082	2879	2250	3061
2024	3107	2940	2226	3116
2025	3168	2999	2238	3168
2026	3207	3041	2223	3202
2027	3292	3097	2220	3250
2028	3347	3158	2242	3303
2029	3397	3203	2228	3342
2030	3459	3260	2247	3390
2031	3560	3325	2248	3447
2032	3573	3357	2271	3473
2033	3645	3423	2257	3530
2034	3691	3481	2265	3581
2035	3757	3510	2254	3603
2036	3793	3567	2281	3653
2037	3848	3639	2278	3716
2038	3893	3692	2276	3764
2039	3954	3734	2278	3799
2040	4045	3786	2282	3841

TABLE 8 | Comparison of diarrhea cases error function of ETS, ARIMA, Holt, and Prophet models.

Model	RMSE	MAE
Prophet	151.62	124.69
ETS	176.21	125.15
ARIMA	192.30	155.94
Holt	408.22	402.38

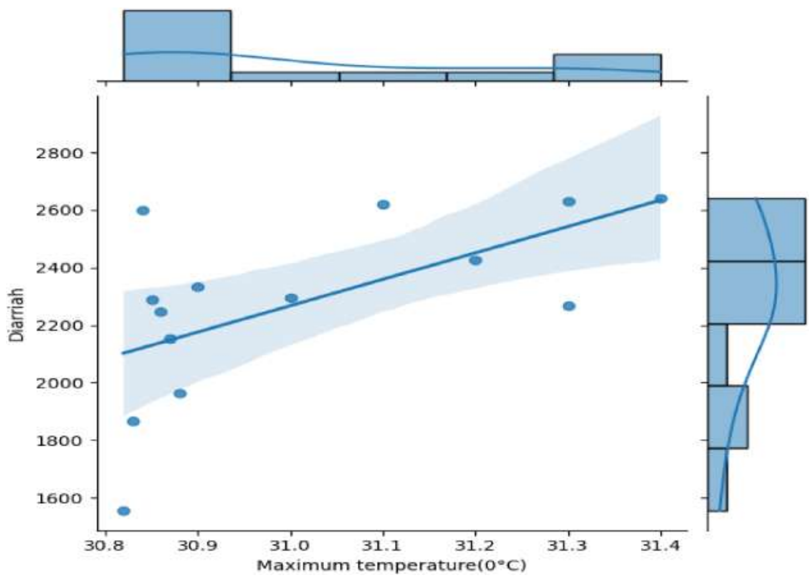

FIGURE 24 | Joint plot showing the relationship between maximum temperature (in °C) and cases of diarrhea.

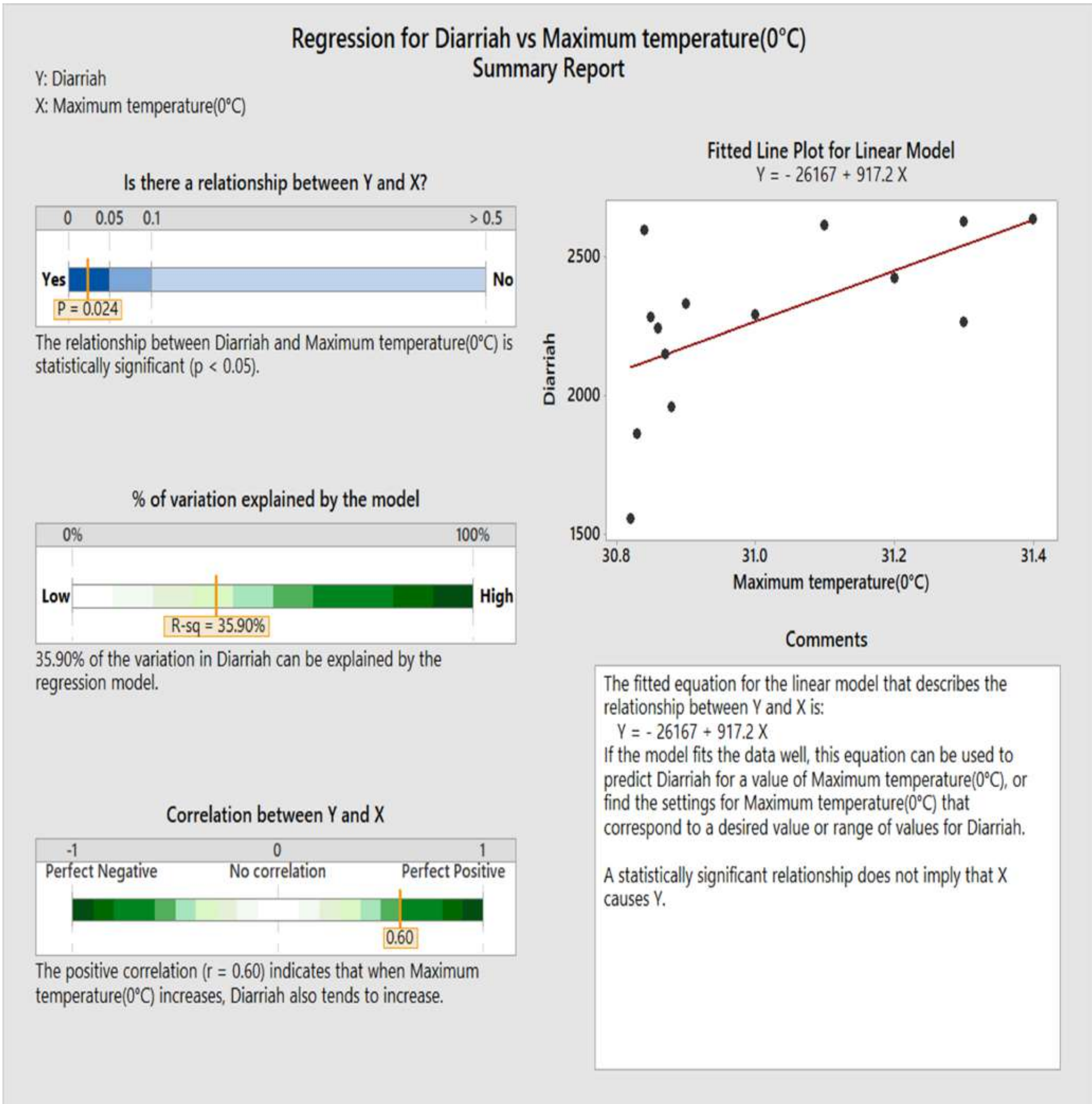

FIGURE 25 | Regression analysis for diarrhea cases versus maximum temperature.

4.9.1 | Comparison With the State of Arts for Forecasting Maximum Temperature in Bangladesh Perspectives

This study represents a comprehensive comparison of maximum temperature forecasts in Bangladesh using advanced time series models such as ETS, Holt, Prophet, and ARIMA models extending projections up to the year 2040 in Table 11. This work, the models consistently predict a maximum temperature of around 31.71˚C to 33.71 from 2020 to 2040 both in the ETS and Holt models. The Prophet model estimates that by 2040, temperatures will reach the Prophet model yielded higher values (32.3˚C–34.3˚C), but moderate error metrics. In addition, this work also forecasted maximum temperatures between 31.61˚C and 32.15˚C up to 2040 by using the ARIMA model. Moreover, the ARIMA model demonstrated superior performance with the lowest RMSE (0.18) and MAE (0.16), confirming its predictive precision as well as reliability.

By comparison, the existing study of Farhana [10] and Sneha et al. [13] is based on average temperatures with their model. The difference with previous works is quite remarkable; this article predicts maximum temperatures using diverse time series models. The existing approaches, for instance that illustrated in the study

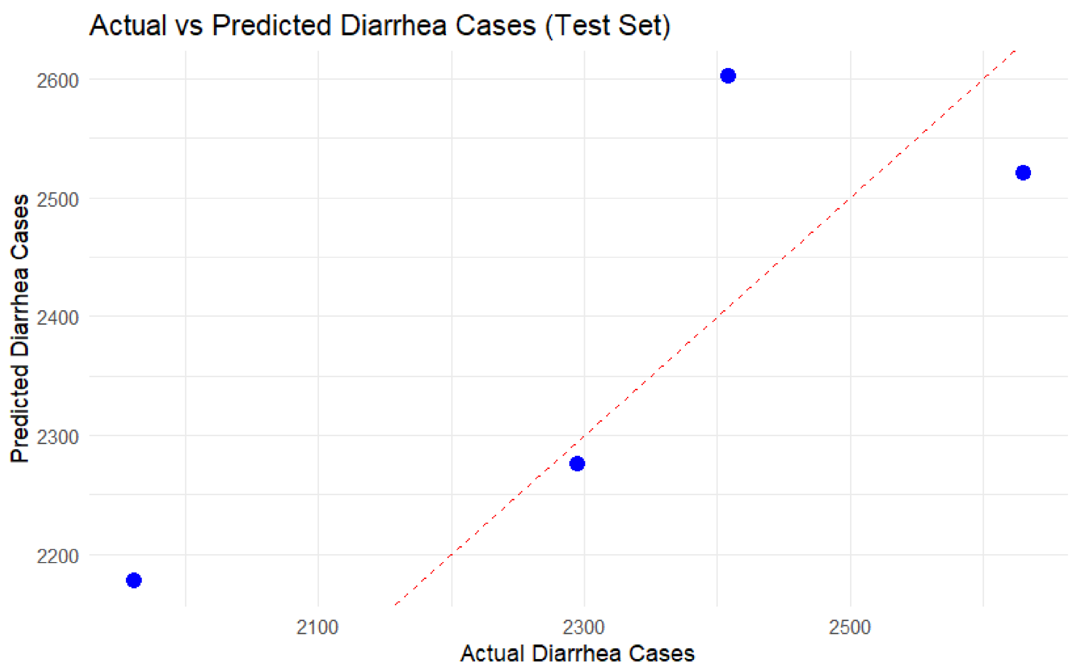

FIGURE 26 | Regression analysis for diarrhea cases versus maximum temperature on test dataset.

TABLE 9 | Linear regression statistical analysis.

Call: lm(formula = Diarrhea_Cases ~ Max_Temp_C, data = train_data)				
Residuals:				
Min	1Q	Median	3Q	Max
−572.49	−41.93	39.07	134.02	454.18
Coefficients:				
	Estimate	Std. Error	*t* value	Pr(>|t|)
(Intercept)—	23028.6	9746.4	−2.363	0.0376 *
Max_Temp_C	816.3	313.8	2.601	0.0246 *
Signif. codes: 0 "0.001" "0.01" "0.05" "." 0.1 "1"				
Residual standard error: 268.2 on 11 degrees of freedom				
Multiple R-squared: 0.3809, Adjusted R-squared: 0.3246				
F-statistic: 6.767 on 1 and 11 DF, *p* value: 0.02464				

TABLE 10 | The estimated error of regression analysis on test dataset.

MAE: 134.67
RMSE: 155.15
R-squared: 0.68

by Farhana [10] predict a time series of average June temperatures using ETS, ARIMA, and SFTS models. The ETS model forecasts temperature at 28.81°C, the ARIMA model forecasts average temperatures at 28.15°C, and the SFTS model has slightly higher values of temperatures at 29.26°C up to 2040. In Sneha et al. [13] study, the polynomial regression models have average predictive temperatures in Bangladesh. The model also forecasts temperatures of 25.62°C in the year 2025, increasing to 25.66°C for each additional 5 years, and a reading of temperature (25.74°C) is reached in 2040.

The key distinction of this study lies in forecasting maximum temperatures and using yearly data, three advanced time series models (ETS, ARIMA, and Holt) and Prophet, whereas previous works by Farhana [10] and Sneha et al. [13] focus solely on average temperatures, often for a single month (e.g., June). This work, ETS, Holt, Prophet, and ARIMA models have been utilized that provide consistent results and more accuracy, forecasting temperatures ranging from 31.39°C to 34.3°C, up to 2040. However, previous work has pointed to remaining steady temperatures. With this approach, the study can provide information that better reflects the present scenario and future climate increases in Bangladesh. The model's better results prove that it is useful in climate adaptation as well as public health planning.

4.9.2 | Comparison With the State of Arts for Maximum Temperature vs. Diarrhea Cases in Bangladesh Perspectives

In this study, the forecasting of diarrhea cases in Bangladesh from 2017 to 2040 using ARIMA, ETS, Holt, and Prophet is compared based on increasing maximum temperatures in Table 12. All models projected an increasing trajectory in both temperatures as well as diarrhea cases. The regression analysis illustrates a statistically considerable positive relationship between maximum temperature and diarrhea cases in Bangladesh, with an R^2 of 0.381 on the training set and 0.68 on the test set, indicating strong predictive capability. According to the ETS, ARIMA, Holt, and Prophet models in 2017, the diarrhea cases recorded at 31.41°C, 31.39°C, 31.4°C, and 32°C were 2561, 2620, 2198, and 2788, respectively. In 2040, Holt and ETS predicted the diarrhea case count of 3786 and 2282 at 33.71°C, and Prophet estimated 3841 cases at 34.3°C. At 32.15°C, ARIMA predicted the highest diarrhea case count of 4045. ETS and Holt both showed a steady rise year after year since they use the same smoothing method. Prophet measured more sensitivity to temperature changes, showing the biggest projected rise.

TABLE 11 | Comparison with the state of the arts for forecasting maximum temperature in Bangladesh perspectives.

Forecasted maximum & average temperature (˚C) (approx.)								
References	This work				Farhana [10]			Sneha Paul [13]
Model ➡ Year ⬇	ETS (Max. Temp. (˚C))	Holt (Max. Temp. (˚C))	Prophet (Max. Temp. (˚C))	ARIMA (Max. Temp. (˚C))	ETS (Avg. Temp. (˚C)) (June)	ARIMA (Avg. Temp. (˚C)) (June)	SFTS (Avg. Temp. (˚C)) (June)	Polynomial Regression (Avg. Temp. (˚C))
2020	31.71	31.71	32.3	31.61	28.81	28.70	29.26	—
2021	31.81	31.81	32.4	31.67	28.81	28.69	29.28	—
2022	31.91	31.91	32.5	31.73	28.81	28.15	29.29	—
2023	32.01	32.01	32.6	31.77	28.81	28.15	29.29	—
2024	32.11	32.11	32.7	31.82	28.81	28.15	29.29	—
2025	32.21	32.21	32.8	31.85	28.81	28.15	29.29	25.62
2026	32.31	32.31	32.9	31.89	28.81	28.15	29.29	—
2027	32.41	32.41	33	31.92	28.81	28.15	29.29	—
2028	32.51	32.51	33.1	31.95	28.81	28.15	29.29	—
2029	32.61	32.61	33.2	31.97	28.81	28.15	29.29	—
2030	32.71	32.71	33.3	32	28.81	28.15	29.29	25.66
2031	32.81	32.81	33.4	32.02	28.81	28.15	29.29	—
2032	32.91	32.91	33.5	32.04	28.81	28.15	29.29	—
2033	33.01	33.01	33.6	32.06	28.81	28.15	29.29	—
2034	33.11	33.11	33.7	32.08	28.81	28.15	29.29	—
2035	33.21	33.21	33.8	32.09	28.81	28.15	29.29	25.70
2036	33.31	33.31	33.9	32.11	28.81	28.15	29.29	—
2037	33.41	33.41	34	32.12	28.81	28.15	29.29	—
2038	33.51	33.51	34.1	32.13	28.81	28.15	29.29	—
2039	33.61	33.61	34.2	32.14	28.81	28.15	29.29	—
2040	33.71	33.71	34.3	32.15	—	—	—	25.74

In case of accuracy, Prophet outweighs all models with the lowest forecast errors (MAE: 124.69, RMSE: 151.62). ETS followed closely (MAE: 125.15, RMSE: 176.21), whereas Holt demonstrated less moderate performance. ARIMA had moderate errors (MAE: 155.94, RMSE: 192.30), indicating limited long-term forecasting capability. Overall, the ARIMA model demonstrated better results than other models at forecasting, as observed by its continued low RMSE and MAE for both maximum temperature as well as diarrhea cases. Even though the Prophet model performs very well for diarrhea predictions, it only exhibits moderate results when predicting maximum temperatures. Thus, ARIMA is more effective and proven to perform well for climate and health forecasting in Bangladesh.

The study is substantial in uniquely identifying a high association between the maximum temperature and the incidence of diarrhea in Bangladesh; a gap that existed in the knowledge of previous researchers. Through effective forecasting techniques, it provides accurate output and serves beneficial insight to health disease analysis. Furthermore, policymakers and healthcare authorities especially can ameliorate readiness and response to extreme heat events by incorporating the suggested forecasting system into the decision-making process in the area of public health. The early warning systems may be supported by real-time forecasts, healthcare resource allocation, and heat-health action plans, such as public warnings, cooling center activation, and vulnerable population protection. Therefore, the outcomes can be utilized in longer-term climate adaptation policy through the support of heat-resilient healthcare infrastructure investments as well as increasing coordination between meteorological and the public health agencies, which will alleviate heat-related health risks in vulnerable locations.

4.10 | Limitations of the Study and Scope of the Further Research

This study, while providing valuable insights, has certain limited health data sources. There are considerable forecasting constraints in prophet and ARIMA models. Furthermore, these statistical models are primarily founded on past history and might not be precise in long-term predictions or in extreme weather fluctuations. Prophet has nonlinear interaction issues that are not multivariate, and ARIMA uses manual parameter tuning, which is less adaptive. Thus, complicated correlations of various climate and health factors might not be well represented during prolonged time intervals. However, this work recommends improvements in future research by expanding multivariate frameworks and addressing data limitations as well as offering a more comprehensive view of health data sources.

5 | Conclusion

The study underscores significant proactive measures to reduce the threats from extreme heat in Bangladesh. It offers exact forecasts through a machine learning model that incorporates influencing public health interventions and policymaking. This

TABLE 12 | Comparison maximum temperature (˚C) versus diarrhea cases in Bangladesh perspectives.

Maximum temperature (˚C) versus diarrhea cases								
References	**This work**							
Model	**ARIMA**		**ETS**		**Holt**		**Prophet**	
Year	**Max. temp. (˚C)**	**Diarrhea cases**	**Max. temp. (˚C)**	**Diarrhea cases**	**Max. temp. (˚C)**	**Diarrhea cases**	**Max. temp. (˚C)**	**Diarrhea cases**
2017	31.39	2620	31.41	2561	31.41	2198	32	2788
2018	31.47	2757	31.51	2628	31.51	2199	32.1	2848
2019	31.55	2814	31.61	2669	31.61	2220	32.2	2881
2020	31.61	2893	31.71	2709	31.71	2202	32.3	2914
2021	31.67	2953	31.81	2784	31.81	2217	32.4	2983
2022	31.73	2997	31.91	2825	31.91	2243	32.5	3016
2023	31.77	3082	32.01	2879	32.01	2250	32.6	3061
2024	31.82	3107	32.11	2940	32.11	2226	32.7	3116
2025	31.85	3168	32.21	2999	32.21	2238	32.8	3168
2026	31.89	3207	32.31	3041	32.31	2223	32.9	3202
2027	31.92	3292	32.41	3097	32.41	2220	33	3250
2028	31.95	3347	32.51	3158	32.51	2242	33.1	3303
2029	31.97	3397	32.61	3203	32.61	2228	33.2	3342
2030	32	3459	32.71	3260	32.71	2247	33.3	3390
2031	32.02	3560	32.81	3325	32.81	2248	33.4	3447
2032	32.04	3573	32.91	3357	32.91	2271	33.5	3473
2033	32.06	3645	33.01	3423	33.01	2257	33.6	3530
2034	32.08	3691	33.11	3481	33.11	2265	33.7	3581
2035	32.09	3757	33.21	3510	33.21	2254	33.8	3603
2036	32.11	3793	33.31	3567	33.31	2281	33.9	3653
2037	32.12	3848	33.41	3639	33.41	2278	34	3716
2038	32.13	3893	33.51	3692	33.51	2276	34.1	3764
2039	32.14	3954	33.61	3734	33.61	2278	34.2	3799
2040	32.15	4045	33.71	3786	33.71	2282	34.3	3841

proposed scheme can forecast maximum temperatures up to the year 2040 and also predict heat-related diseases such as diarrhea cases up to 2040 for future public health scenarios. This work delineates four forecasting models such as ARIMA, ETS, Holt, and Prophet for maximum temperature and diarrhea cases commencing from 2017 till 2040 in Bangladesh. This study demonstrated a statistically significant correlation between maximum temperature and diarrhea cases in Bangladesh, supported by regression analysis with an R^2 of 0.381 for the training set and 0.68 for the test set. In addition, it also exhibits that the Pearson correlation coefficient of 0.60 represents a moderate positive linear relationship between cases of diarrhea and maximum temperature (in ˚C) in Bangladesh. In 2017, all models measured comparable estimates, while in 2040, ETS and Holt predicted the number of cases respectively 3786 and 2282 at 33.71˚C, and Prophet followed closely with 3841 cases at 34.3˚C. However, ARIMA predicted the highest diarrhea cases (4045) at 32.15˚C. Among the models, ARIMA demonstrated the best overall performance, achieving moderate error margins for diarrhea cases (MAE: 155.94, RMSE: 192.30) and the lowest errors for maximum temperature (MAE: 0.16, RMSE: 0.18). The Prophet model excelled in forecasting diarrhea cases, yet it showed only moderate accuracy for temperature predictions. Therefore, ARIMA emerges as the most effective model to confirm its reliability on forecasting maximum temperature and diarrhea cases.

To recapitulate, this article forecasts the very satisfactory accuracy of maximum temperatures up to 2040 linked to heat-related diseases in public health scenarios, demonstrating the best accurate predictions for diarrhea case management. Thus, this work outclasses all the existing approaches and presents consistent forecasting of maximum temperature with improved accuracy. Moreover, the research contributes to the significant gap in earlier studies by demonstrating a close association between maximum temperature and diarrhea in Bangladesh. However, this work recommends improvements in future research by addressing reliable sources of data. Therefore, the findings of this study can contribute to global efforts on public health to mitigate the adverse impacts of climate change and provide useful insights for policymakers which can develop targeted interventions to protect vulnerable populations, enhance healthcare infrastructure, and promote climate-resilient development in Bangladesh.

Author Contributions

Conception and design: Tanvir Ehsan, Farhana Akter Bina, Gazi Md. Habibul Bashar, Liton Chandra Paul; analysis and interpretation of the data: Tanvir Ehsan, Farhana Akter Bina, Gazi Md. Habibul Bashar, Liton Chandra Paul; drafting of the paper: Tanvir Ehsan, Farhana Akter Bina, Gazi Md. Habibul Bashar, Liton Chandra Paul; revising it critically for intellectual content: Tanvir Ehsan, Farhana Akter Bina, Gazi Md. Habibul Bashar, Liton Chandra Paul; and final approval of the version to be published: Tanvir Ehsan, Farhana Akter Bina, Gazi Md. Habibul Bashar, Liton Chandra Paul. All the authors agree to be accountable for all aspects of the work.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. M. M. Hamed, A. A. J. A. Hasani, M. S. Nashwan, Z. Sa'adi, and S. Shahid, "Assessing the Growing Threat of Heat Stress in the North Africa and Arabian Peninsula Region Connected to Climate Change," *Journal of Cleaner Production* 447 (2024): 141639.

2. G. Miloshevich, D. Lucente, P. Yiou, and F. Bouchet, "Extreme Heat Wave Sampling and Prediction With Analog Markov Chain and Comparisons With Deep Learning," *Environmental Data Science* 3 (2024): e9.

3. S. P. Shuvo, S. Pal Shuvo, J. MdAshikuzzaman, et al., "Hybrid Noise Reduction-Based Data-Driven Modeling of Relative Humidity in Khulna, Bangladesh," *Heliyon* 10, no. 16 (2024): e36290.

4. M. Mehrin, F. T. Zuhra, and M. M. Rahman, "Heatwave Induced Health Vulnerability Assessment in Bangladesh," *Research Square (Research Square)* 3, no. 1 (2024): 1–18.

5. Z. J. Khan, *A Machine Learning Analysis of Climate Change & Human Health Projections in Pakistan* (Pakistan Institute of Development Economics, 2024).

6. M. Y. Shams, Z. Tarek, A. M. Elshewey, M. Hany, A. Darwish, and A. E. Hassanien, "A Machine Learning-Based Model for Predicting Temperature Under the Effects of Climate Change," *Studies in Big Data* (2023): 61–81.

7. N. Narkhede, R. Chattopadhyay, S. Lekshmi, P. Guhathakurta, N. Kumar, and M. Mohapatra, "An Empirical Model-Based Framework for Operational Monitoring and Prediction of Heatwaves Based on Temperature Data," *Modeling Earth Systems and Environment* 8, no. 4 (2022): 5665–5682.

8. M. Mannan, F. Akter, M. B. Rashid, et al., "An Analytical Study of Heat Wave Occurred Over Bangladesh During Pre- Monsoon of 2021," *Dewdrop* 8 (2022): 152–160.

9. H. M. Imran, A. Hossain, A. K. M. S. Islam, et al., "Impact of Land Cover Changes on Land Surface Temperature and Human Thermal Comfort in Dhaka City of Bangladesh," *Earth Systems and Environment* 5, no. 3 (2021): 667–693.

10. F. A. Bina, "Forecasting of Temperature in Bangladesh by Using Sliced Functional Time Series (Sfts) Approach," *International Journal of Advanced Scientific and Technical Research* 5, no. 10 (2020): 51–66.

11. S. Karmakar and M. K. Das, "On the Heat Waves in Bangladesh, Their Trends and Associated Large Scale Tropospheric Conditions," *Journal of Engineering Science* 11, no. 1 (2020): 19–36.

12. M. Z. Ali, G. Carlile, and M. Giasuddin, "Impact of Global Climate Change on Livestock Health: Bangladesh Perspective," *Open Veterinary Journal* 10, no. 2 (2020): 178–188.

13. S. Paul and S. Roy, "Forecasting the Average Temperature Rise in Bangladesh: A Time Series Analysis," *Journal of Engineering Science* 11, no. 1 (2020): 83–91.

14. S. K. Sarkar, R. R. Rudra, S. Talukdar, et al., "Future Groundwater Potential Mapping Using Machine Learning Algorithms and Climate Change Scenarios in Bangladesh," *Scientific Reports* 14, no. 1 (2024): 10328.

15. S. Ullah, M. Abbas, and X. Qiao, "Impact Assessment of Land-Use Alteration on Land Surface Temperature in Kabul Using Machine Learning Algorithm," *Journal of Spatial Science* (2024): 1–23.

16. A. Addas, "Machine Learning Techniques to Map the Impact of Urban Heat Island: Investigating the City of Jeddah," *Land* 12, no. 6 (2023): 1159.

17. A. A. Kafy, N. N. Dey, A. Al Rakib, Z. A. Rahaman, N. M. R. Nasher, and A. Bhatt, "Modeling the Relationship Between Land Use/Land Cover and Land Surface Temperature in Dhaka, Bangladesh Using CA-ANN Algorithm," *Environmental Challenges* 4 (2021): 100190.

18. A. A. Kafy, A. Al-Faisal, A. Al Rakib, et al., "The Operational Role of Remote Sensing in Assessing and Predicting Land Use/Land Cover and Seasonal Land Surface Temperature Using Machine Learning Algorithms in Rajshahi, Bangladesh," *Applied Geomatics* 13, no. 4 (2021): 793–816.

19. S. Karmakar, M. K. Das, and M. R. Alam, "Study on Heat Waves and Associated Large-Scale Circulations in Bangladesh," in *National Oceanographic and Maritime Institute (NOAMI), final Report* (National Oceanographic And Maritime Institute (NOAMI), 2019).

20. X. Xian, L. Wang, X. Wu, et al., "Comparison of SARIMA Model, Holt-Winters Model and ETS Model in Predicting the Incidence of Foodborne Disease," *BMC Infectious Diseases* 23, no. 1 (2023): 803.

21. S. Naher, F. Rabbi, M. M. Hossain, R. Banik, S. Pervez, and A. B. Boitchi, "Forecasting the Incidence of Dengue in Bangladesh—Application of Time Series Model," *Health Science Reports* 5, no. 4 (2022): e666.

22. M. A. Hasan, I. R. Abubakar, S. M. Rahman, Y. A. Aina, M. M. Islam Chowdhury, and A. N. Khondaker, "The Synergy Between Climate Change Policies and National Development Goals: Implications for Sustainability," *Journal of Cleaner Production* 249 (2019): 119369.

23. Z. Z. Oo and S. Phyu, "Time Series Prediction Based on Facebook Prophet: A Case Study, Temperature Forecasting in Myintkyina," *International Journal of Applied Mathematics Electronics and Computers* 8, no. 4 (2020): 263–267.

24. I. J. A. Pesigan and S. F. Cheung, "Monte Carlo Confidence Intervals for the Indirect Effect With Missing Data," *Behavior Research Methods* 56, no. 3 (2024): 1678–1696.

25. W. Cui, Z. Sun, H. Ma, and S. Wu, "The Correlation Analysis of Atmospheric Model Accuracy Based on the Pearson Correlation Criterion," *IOP Conference Series: Materials Science and Engineering* 780, no. 3 (2020): 32045.

26. Bangladesh Meteorological Department (BMD) accessed June 18, 2025, https://www.bmd.gov.bd/.

27. Directorate General of Health Services (DGHS) accessed June 18, 2025, https://dghs.gov.bd/.

28. Climate Change and Health in Bangladesh, *People's Republic of Bangladesh Health Services Division, Ministry of Health and Family Welfare* (People's Republic of Bangladesh Health Services Division, Ministry of Health and Family Welfare, 2022), https://dghs.portal.gov.bd/sites/default/files/files/dghs.portal.gov.bd/page/a5b182fd_dea0_4512_b0cd_1c9bfe29b436/2022-03-03-10-17-072edab52fdfb00922a587f0203f0197.pdf.